The Drug Main Effect in Aggregated N-of-1 Designs under the Refined Process: ADEMP Working Protocol for the Rebuild of Paper 04

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Manuscript. `analysis/report/04-treatment-main-effect/report.Rmd` (to be rebuilt).

Protocol history. This file replaces the working protocol of 2026-05-07, which is retained unchanged as `analysis/scripts/treatment-main-effect/00-ademp-protocol-v1.md` (byte-identical copy, checked with `cmp`). Version 1 describes the study behind the current manuscript and its driver `production-run.R`; it is kept as the record of that study.

Purpose. This working protocol sets out a new simulation study of the treatment (drug) main effect β_D in aggregated N-of-1 designs, to replace the study of the current paper 04. It is a planning document: no code has been written for it and nothing has been simulated. Section 8 lists the decisions that must be settled before code is written. The study is exploratory. Its criteria may be revised as results come in, every revision will be recorded as an amendment with its reason, and its findings should be confirmed in an independent study. The structure follows ADEMP: aims, data-generating mechanisms, estimands, methods and performance measures (Morris, White and Crowther [1]).

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here and are not part of the canonical set.

Table 1: Notation index for symbols used in this protocol

Symbol	Meaning	Status
i, t	participant and visit indices; $t = 0$ is baseline	canonical
w_t	calendar week of visit t	local (`docs/40`)
N	total participants across all paths or arms	canonical
P	number of randomization paths (4 Hybrid, 2 OL+BDC, 2 CO)	canonical
Y_{it}	symptom score (CAPS points; code column `Sx`)	canonical
BL_i	baseline level	canonical
BR_{it}, PB_{it}, TV_{it}	drug, placebo-belief and natural-course responses	canonical
η_t	design expectancy: 1 open label, 0.5 blinded, 0 at baseline (code `De`)	canonical
D_{it}, $D_{bc,it}$	binary drug state; exposure-decayed indicator, 0 before any exposure (code `Dbc`)	canonical
t_{od}, t_{sd}, t_{pb}	time on drug, time since discontinuation, time under expectancy	canonical
$t_{1/2}$, $t^a_{1/2}$	true carryover half-life; half-life assumed by an analysis (weeks)	canonical; local (`docs/45`)
$G(t; m, d, r)$	modified Gompertz curve	local (`docs/40`)
m_{BR}, m_{TV}, m_{PB}	Gompertz ceilings of the three components (published 10.986, 6.506, 6.506)	local
f_m	drug-effect multiplier, $m_{BR}/10.986$	local
q_{it}	unit drug-response profile: the BR mean of participant i's path divided by m_{BR}, anchored carryover included	local
v_i	participant-specific deviation of the drug-response ceiling	local
τ^2	variance of v_i, the participant-by-treatment variance (σ^2_v of `docs/51`)	local
β_D	drug main effect, the primary estimand	canonical
ρ, c_1	AR(1) serial correlation per week (0.7); cross-component correlation (0.2)	canonical; local (`docs/40`)

Symbol	Meaning	Status
σ_{BR}	SD of BR (8)	canonical
$\sigma_{\varepsilon,0}$, r_{BL}	SD of measurement error added to the observed baseline; its reliability $\sigma^2_{BL}/(\sigma^2_{BL}+\sigma^2_{\varepsilon,0})$	local
$\sigma_{\varepsilon,\mathrm{OL}}$, $\sigma_{\varepsilon,\mathrm{bl}}$	SD of measurement error at open-label and blinded visits	local (`docs/46`)
κ	mean model SE over empirical SD	canonical
α, π	nominal level (0.05); power	canonical
n_{reps}	Monte Carlo replicates per cell	canonical

Two collisions are avoided deliberately. `docs/37` uses τ^2 for the residual variance of the paired contrast given the biomarker; that quantity is written $\sigma^2_{\Delta|B}$ here, and τ^2 is reserved for the participant-by-treatment variance, as in the literature on series of N-of-1 trials [2, 3]. The canonical δ (standardized bias) and θ (generic interaction coefficient) are not used for the drug effect.

Sign convention (NOTATION.md, rule 5). The components are non-negative reductions in symptom severity, so an increase in any of them lowers Y. The drug main effect β_D is therefore negative when the drug helps; m_{BR}, f_m and τ are non-negative. Bias is reported on the β_D scale, so a positive bias is a bias toward no effect.

Units. Carryover half-lives are in **weeks** throughout, matching the refined process and the interaction manuscripts. This departs from `NOTATION.md`, Part 1, "Units", which assigns days to the main-effect manuscript; that paragraph must be revised when the rebuilt paper is written (1 week = 7 days). Effect sizes are given in CAPS points and in SD units (Section 3.5).

Evidence labels. As in `docs/53`: *quoted*, *inspected* (read from source), *derived* (follows algebraically), *verified* (computed by running code; here only arithmetic checks in the session scratchpad, `estimand-check.R`, not a project script), *inferred* and *assumed*.

Glossary.

- **The refined process.** The recommended data-generating process (DGP) of paper 01, Table `tab:spec`; "Refined" in tables. **The published process**: Hendrickson's code at commit 58b32a9 (`NOTATION.md`, "Named data-generating processes").
- **Hybrid.** The Hendrickson design: open label to week 8, blinded discontinuation in weeks 8 to 12, a blinded two-period crossover in weeks 12 to 20; visits at weeks 4, 8, 9, 10, 11, 12, 16, 20 [4].
- **Multi-crossover.** The design of the current paper 04: eight one-week periods, four on drug and four on placebo in a random order per participant. It is not the Hybrid design.

- **Drug-expectancy confounding.** In open-label phases every visit is on drug at full expectancy, so the drug state and η_t are correlated (0.588 over the nine Hybrid rows; `compendium-mathematics.tex`, Section "The published Hybrid construct", verified there). An analysis that does not model η_t credits part of PB to the drug.
- **Heterogeneous null.** A cell with $f_m = 0$ and $\tau > 0$: the mean drug effect is zero but individual effects are not.
- **Size-adjusted power, shortfall, minimax regret.** As in `docs/46` and `docs/47`: power at the 5% quantile of the analysis's own null p-values; an analysis's shortfall in a cell is its distance from the best surviving analysis there; the selected analysis minimizes its largest shortfall.
- **Oracle analysis.** An analysis that uses information the trial would not have (a true half-life, the true Gompertz shape, an error-free baseline). Reported for reference; never selectable.

1. Why the rebuild

The current paper's simulation does not study the design, the process or the inference that the paper describes. Seven problems were found (inspected, 2026-10-08):

1. **The design is not the Hybrid design.** The driver generates a randomized 4 + 4 multi-crossover of one-week periods (`production-run.R`, lines 80 to 85: `seq_base` of four 1s and four 0s, permuted by `sample()`), while the abstract and Results call the N-of-1 arm "hybrid" (`report.Rmd`, line 8; lines 1216 to 1221). Methods describes the multi-crossover correctly (lines 515 to 517).
2. **The outcome has no placebo response, natural course, expectancy or onset.** The outcome is `baseline + Dbc * true_effect + carryover_adj + eps` (`production-run.R`, line 103). There is no PB, no TV, no η_t and no Gompertz onset, so the drug-expectancy confounding of the open-label phase never arises, and the analysis's linear time term is fitted to a mean with no trend.
3. **No treatment-effect heterogeneity.** Every participant has the same effect (`true_effect`, line 103); the paper says that between-patient heterogeneity lies "purely" in baseline frequency (`report.Rmd`, lines 1362 to 1374), although its own literature review rests on τ^2 (lines 153 to 252).
4. **Carryover exclusion uses oracle information.** The analysis drops the rows flagged `is_carryover`, a column the DGP writes (`production-run.R`, lines 110 and 120 to 127). A trial does not know which visits carry residual drug effect.
5. **Carryover is truncated and on a day scale.** Carryover reaches only the period immediately after a drug period (lines 94 to 101), and half-lives are in days (lines 53 and 95), off the scale of every other paper in the compendium.

6. **The inference labeled Kenward-Roger is not.** The test uses the `nlme` table's degrees of freedom (lines 172 to 178), which are the containment (between-within) degrees of freedom, not Kenward-Roger and not Satterthwaite, although the header (line 13) and Methods (`report.Rmd`, lines 564 to 568) say otherwise; the reduced model fitted for the correction (lines 159 to 170) is never used.
7. **The randomized-trial covariate is measured without error.** The RCT endpoint is generated from, and adjusted for, the participant's true intercept (`production-run.R`, lines 197 to 204 and 235). An ANCOVA on an error-free covariate removes all between-participant baseline variance, which favors the RCT in a way no trial can reproduce. The RCT arm also has no placebo response or natural course.

The compendium's own documents had already recorded parts of this (`compendium-mathematics.tex`, Section "The treatment main-effect track", notes of 2026-10-08), and the remediation plan (`analysis/report/04-treatment-main-effect/00-remediati` Problems 4 and 5) asked for a correct small-sample test and an estimand defined under the Gompertz process. Neither can be repaired inside the current DGP, because the DGP lacks the features that make the estimand nontrivial. Hence a rebuild rather than a revision.

2. Aims (A)

Primary aim. Estimate the size and power of tests of the drug main effect β_D in the Hybrid design under the refined process, and compare them with the OL+BDC design, the two-period crossover (CO) and a parallel-group randomized controlled trial (RCT), all simulated from the same component model at the same total N.

Secondary aims.

1. Estimate the participant-by-treatment variance τ^2, and assess how well each design recovers individual drug effects.
2. Recommend an analysis for the drug main effect in each design, by the working decision rule of Section 6.4.
3. Quantify, in closed form first, how much of each design's power comes from randomized between-path contrasts and how much from modeled within-participant contrasts, and how much bias the drug-expectancy confounding introduces into each analysis.

What the study does not aim at. The biomarker interaction (papers 01, 02 and the stress test of `docs/46`); optimal design (`docs/39`); the comparison of response-curve families (paper 07).

Prior work. Appendix A reviews the published Monte Carlo and closed-form studies of the treatment main effect in N-of-1, aggregated N-of-1, multi-crossover

and single-case designs, compares their data-generating mechanisms and analyses with this protocol, and lists the proposals that follow from that review (Section 8, items 12 to 20).

3. Data-generating mechanisms (D)

3.1 The refined process

The base DGP is the refined process of paper 01 (Table `tab:spec`; Appendix B), with the biomarker uncoupled ($c_{bm} = 0$) in the base cell. For participant i on path p,

$$Y_{i0} = \mathrm{BL}_i, \qquad Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it}) - v_i\, q_{it} + \varepsilon_{it}^{\mathrm{me}}, \quad t \geq 1.$$

- **Components and means.** $E(TV_{it}) = G(w_t; m_{TV}, 5, 0.35)$ on calendar time; $E(PB_{it}) = \eta_t\, G(t_{pb}; m_{PB}, 5, 0.35)$; $E(BR_{it}) = G(t_{od}; m_{BR}, 5, 0.42)$ on drug, with the published ceilings $m_{TV} = m_{PB} = 6.506$ and $m_{BR} = 10.986$ unless a cell varies them. BL_i has mean 83.07 and SD 18.48 and is independent of the components (`docs/40`, Section 3).
- **Expectancy.** η_t scales both the mean and the SD of PB (SD $10\eta_t$); this is the only route by which blinding enters the data (`docs/40`, Section 3).
- **Covariance.** Separable AR(1), $K \otimes A$ with $A_{ts} = \rho^{|w_t - w_s|}$, $\rho = 0.7$ per week, and cross-component correlation $c_1 \rho^{|w_t - w_s|}$, $c_1 = 0.2$; SDs 10 (TV), $10\eta_t$ (PB) and 8 (BR) at every visit, on and off drug (paper 01, "Why separability").
- **Anchored carryover.** Off drug after exposure, $E(BR_{it}) = \mu_{BR,\mathrm{last}}\, 2^{-t_{sd}/t_{1/2}}$, decayed once from the last on-drug mean, not by the published compounding recursion (`docs/53`; paper 01, Appendix B.3). The same decay applies to the drug-response profile q_{it}.
- **Half-life.** Pharmacological range $t_{1/2} \in \{0.1, 0.2\}$ week (17 to 34 hours, about 7 to 15 plasma half-lives of prazosin; paper 01, "Pharmacological grounding"), base 0.2. A **behavioral carryover scenario** at $t_{1/2} = 1$ week is simulated and labeled as such everywhere it is reported: it is not a pharmacological claim about prazosin.
- **Re-exposure.** As in the published process, t_{od} restarts at zero when the drug is resumed (justified by re-titration of prazosin). Combined with anchored carryover, this gives a downward step in the mean at the first re-exposure visit when the residual exceeds the restarted curve (paper 01, Appendix B.3).
- **Validity.** The generator stops on a correlation matrix that is not positive definite (paper 01, Appendix B.2). With $c_{bm} = 0$ no cell is near a ceiling; the stop is kept for the cell with $c_{bm} > 0$ (Section 3.6).

- **Allocation.** Each participant is assigned to a path, or to an RCT arm, uniformly at random in every trial, redrawn if a path or arm receives fewer than 2 participants (the rule of `11-strawman-stress-test.R`, `RERUN-anchored.md`, Section 1). N is the total (NOTATION.md, rule 4).

3.2 New element: a random drug-effect slope

The refined process has no participant-by-treatment variance: the BR component varies between participants with SD 8 equally on and off drug, so its person-level part cancels in an on-minus-off contrast (`docs/51`, Feature 8; derived). Treatment-effect heterogeneity is added as a participant-specific ceiling,

$$v_i \sim N(0, \tau^2), \quad \text{independent of everything else,} \qquad q_{it} = E(BR_{it})/m_{BR},$$

that is, q_{it} is the drug-response mean of participant i's path computed with unit ceiling, including onset and anchored carryover; it does not depend on m_{BR} and is defined when $m_{BR} = 0$. Participant i's expected drug response is then $(m_{BR}+v_i)\, q_{it}$: heterogeneity follows the drug effect, is zero before exposure and decays with the drug effect, as `docs/51` recommends ("let the drug-response variance follow the drug effect"). τ is set in CAPS points, independent of m_{BR}, so that heterogeneous nulls ($m_{BR} = 0$, $\tau > 0$) exist. A normal v_i allows a participant to be harmed when $\tau > m_{BR}$; that is intended in the heterogeneous null. Whether v_i should instead scale the binary on-drug state, or be correlated with BL_i, is an open decision (Section 8, item 9).

3.3 Designs

Table 2: Designs, randomization paths, schedules and expectancy

Design	P	Visit weeks (after baseline)	η_t	Paths (drug state by visit)
Hybrid	4	4, 8, 9, 10, 11, 12, 16, 20	1, 1, then 0.5	11110010, 11110001, 11100010, 11100001
OL+BDC	2	4, 8, 12, 16, 17, 18, 19, 20	1 to week 16, then 0.5	11111100, 11111000
CO	2	every 2.5 weeks to week 20	0.5	11110000, 00001111

Design	P	Visit weeks (after baseline)	η_t	Paths (drug state by visit)
RCT (parallel)	2 arms	4, 8, 9, 10, 11, 12, 16, 20	0.5	11111111 or 00000000
Multi-crossover (optional)	random	eight weekly visits	0.5	random 4 + 4 per participant

The first three rows are the designs of `analysis/scripts/quick-sim/hendrickson-problems/04-power-simulation.R`, lines 74 to 83 (inspected). All have 8 post-baseline visits over 20 weeks [4].

The RCT arm. A parallel-group comparator of the kind that motivated the Hybrid design [13], double-blind throughout ($\eta_t = 0.5$ from week 0), so PB and TV follow the same curves as in the N-of-1 designs and are identical in the two arms. The active arm has $t_{od} = w_t$; the placebo arm has BR mean 0. Two properties need stating:

- **Inherited BR noise.** Under the refined process the placebo arm still carries BR noise with SD 8, as an unexposed crossover path does. This is the convention of the published construct, not pharmacology (`docs/51`, Feature 8); it adds 64 to the variance of every RCT measurement. Whether to keep it is an open decision (Section 8, item 10).
- **Baseline measured with error.** The analysis sees $Y^*_{i0} = \mathrm{BL}_i + \varepsilon_{i0}$, $\varepsilon_{i0} \sim N(0, \sigma^2_{\varepsilon,0})$, at reliability r_{BL}. Published test-retest reliabilities of the CAPS are high but below 1 [14], so an error-free baseline is the optimistic limit. The same observed baseline is used in the N-of-1 designs, where the baseline row then has its own noise, which also removes the degenerate noise-free baseline row noted in `docs/41`, Proposition 5. The default and the grid are an open decision (Section 8, item 7).

The multi-crossover. Under the refined process with re-titration, a single on-drug week reaches 3.1% of the ceiling (0.34 points at the published m_{BR}; verified, arithmetic), so the design of the current paper cannot express the drug effect of a titrated drug except through runs of consecutive on-drug weeks. It is offered only as an optional bridge to the current paper (Section 8, item 5).

3.4 Factors and grids

Table 3: Data-generating factors, levels and base values

Factor	Levels	Base
Drug effect $f_m = m_{BR}/10.986$	0, 0.2, 0.4, 0.6, 0.8, 1.0	curve; 0.4 in stress cells
Heterogeneity τ (points)	0, 2.75, 5.49 ($\tau/10.986$ = 0, 0.25, 0.5)	0
Total N	35, 70, 150	70
Carryover $t_{1/2}$ (weeks)	0, 0.1, 0.2 (pharmacological); 1.0 (behavioral, labeled)	0.2
Decay form	exponential; Weibull shape 0.5 and 2 at $t_{1/2} = 1$	exponential
Natural course m_{TV}	$\times 0$, $\times 1$, $\times 2$	$\times 1$ (6.506)
Placebo response m_{PB}	$\times 0$, $\times 1$, $\times 2$	$\times 1$ (6.506)
Dropout	none; 20% monotone MCAR; 20% monotone MAR on $Y_{i,t-1}$	none
Phase measurement error	none; $\sigma_{\varepsilon,\mathrm{OL}} = 3$, $\sigma_{\varepsilon,\mathrm{bl}} = 6$	none
Baseline reliability r_{BL}	1; 0.8 ($\sigma_{\varepsilon,0} = 9.24$)	open (Section 8)

The dropout and measurement-error mechanisms are those of the options patch and the stress test (`analysis/patches/orig-to-refined/ README.md`, options table; `docs/46`, Section 5).

3.5 Effect size in outcome and SD units

The drug-effect axis is defined on the DGP, by the ceiling m_{BR}, because each design's estimand takes a different value for the same drug (Section 4). For an SD scale we propose the SD of a blinded post-baseline score conditional on baseline,

$$\sigma^2_{Y|\mathrm{BL}} = \sigma^2_{TV} + (10\eta_t)^2 + \sigma^2_{BR} + 2c_1(10\sigma_{TV}\eta_t + \sigma_{TV}\sigma_{BR} + 10\eta_t\sigma_{BR}) = 257$$

at $\eta_t = 0.5$, so $\sigma_{Y|\mathrm{BL}} = 16.03$ (derived; verified, arithmetic; the same value as `docs/41`, Proposition 3, because marginal variances do not depend on the serial structure). The unconditional SD is $\sqrt{598.5} = 24.46$.

Table 4: Drug-effect grid in multiplier, outcome and SD units

f_m	0	0.2	0.4	0.6	0.8	1.0
m_{BR} (points)	0	2.20	4.39	6.59	8.79	10.99
$m_{BR}/\sigma_{Y\|\mathrm{BL}}$	0	0.14	0.27	0.41	0.55	0.69

The grid was chosen to keep the N-of-1 designs off the power ceiling. That rests on rough derived values (an RCT SE near 3.8 points at $N = 70$ with an error-free baseline, Section 5.1; a Hybrid SE near 1 point, taken from the compound-symmetry results of `docs/40`, Section 9) and must be checked against the closed form of Section 5.1 before the run.

3.6 Draft cell table

Every block varies one factor from the base cell (Hybrid, OL+BDC, CO and RCT; refined process; $N = 70$; $t_{1/2} = 0.2$; $\tau = 0$; no dropout or measurement error). Stress blocks have a null twin ($f_m = 0$) and an alternative at $f_m = 0.4$. The RCT is not simulated in blocks whose factor does not reach it.

Table 5: Draft cell table: blocks, levels and cell counts

Block	Factor varied	Levels	f_m	Designs	Cells (null / alt)
1	power curves	base	0 to 1.0	4	24 (4 / 20)
2	heterogeneity	$\tau = 2.75$, 5.49	0, 0.4	4	16 (8 / 8)
3	sample size	$N = 35$, 150	0, 0.4	4	16 (8 / 8)
4	carryover	$t_{1/2} = 0$; 1.0; Weibull 0.5 and 2 at 1.0	0, 0.4	3 (no RCT)	24 (12 / 12)
5	natural course	$m_{TV} \times 0$, $\times 2$	0, 0.4	4	16 (8 / 8)
6	placebo response	$m_{PB} \times 0$, $\times 2$	0, 0.4	4	16 (8 / 8)
7	dropout	MCAR, MAR (20%)	0, 0.4	4	16 (8 / 8)

Block	Factor varied	Levels	f_m	Designs	Cells (null / alt)
8	phase measurement error	3 / 6	0, 0.4	4	8 (4 / 4)
9	baseline error	$r_{BL} = 0.8$	0, 0.4	4	8 (4 / 4)
Total					144 (64 / 80)

Optional cells, not counted above: $t_{1/2} = 0.1$ (expected to be indistinguishable from 0.2 at weekly visits); explained heterogeneity through covariance moderation, $c_{bm} = 0.30$, $\tau = 0$ (the biomarker then induces a drug-effect SD of $c_{bm}\sigma_{BR} = 2.4$ points on drug; derived from `docs/40`, Section 5); and the multicrossover in blocks 1 and 4.

Misspecified analysis half-life needs no new data: in the cells of block 1 and the $t_{1/2} = 1$ cells of block 4, the D_{bc}-based analyses are also fitted at $t^a_{1/2} \in \{0, 0.5, 1\}$.

Replicates and MCSE targets. Two options (Section 8, item 6):

Table 6: Replicate options with Monte Carlo standard error targets

Option	Null cells	Alternative cells	MCSE of size at 0.05	Largest MCSE of power	MCSE of coverage at 0.95
R1	2,000	1,000	0.0049	0.016	0.0049 (null), 0.0069 (alt)
R2 (as `docs/46`)	1,000	500	0.0069	0.022	0.0069 (null), 0.0097 (alt)

R1 gives 128,000 null and 80,000 alternative trials (208,000), of which about 165,000 are N-of-1 trials; R2 halves both. The MCSE of bias is $\mathrm{SD}(\widehat{\beta}_D)/\sqrt{n_{\mathrm{reps}}}$, about 0.03 points for an empirical SD of 1 at $n_{\mathrm{reps}} = 1{,}000$.

4. Estimands (E)

4.1 Candidate definitions of β_D

Under the refined process the drug effect is not one number: it rises on the Gompertz curve, restarts at re-exposure and decays after discontinuation, and the open-label visits add full expectancy. Every analysis with a single drug coefficient targets some weighted average of a drug response that ranges from 4.3 to 10.2 points across the Hybrid visits (`docs/40`, Section 9). Three definitions were considered. Values are at the published ceiling, $t_{1/2} = 0$, equal path weights (verified, arithmetic, `estimand-check.R`):

1. **Full-effect contrast**, $\beta_D^{\text{full}} = -m_{BR}$: the reduction after long exposure, 10.99 points. Design-free, but no design observes it: no Hybrid or crossover visit is past 10 weeks of continuous exposure, and the re-exposure visits reach 39% of the ceiling. Only an analysis that knows the Gompertz shape can target it (the oracle analysis OP, Section 5); every other analysis is biased toward zero by construction, which makes bias uninformative about the analyses.
2. **Contrast at the design's time on drug**, $\beta_D^{\text{exp}} = -[\overline{E(BR)}_{\text{on}} - \overline{E(BR)}_{\text{off}}]$, the on-minus-off difference in the mean drug response over the design's own visits, averaged over participants, net of PB and TV by definition. It is what the drug did at the times the design exposed people. Values: Hybrid −7.22 (paths 1 and 2, −7.55; paths 3 and 4, −6.89); OL+BDC −9.35; CO −6.70; RCT at week 20, −10.97. Under carryover the off-drug mean is positive and the value shrinks; under expectancy the definition excludes PB, so an analysis that credits open-label expectancy to the drug is biased away from zero.
3. **Blinded on-minus-off contrast at matched expectancy**, β_D^{bl}: the same contrast restricted to blinded visits. Hybrid −7.56 (paths 1 and 2, −8.09; 3 and 4, −7.04); undefined for OL+BDC except at its two or three blinded visits. Its randomized version, the equal-weight average of the between-path contrasts at weeks 10, 16 and 20, is −6.25 in the Hybrid design, the probability limit of the visit-factor analysis (`docs/40`, Section 9; `compendium-mathematics.tex`, Equation `eq:bp-fwl`). Matching expectancy removes the PB step but not the natural course: the blinded within-path contrast without time adjustment has a null bias of +0.37 points, against +1.29 for the contrast over all visits (verified, arithmetic).

Implications under Gompertz onset. Definitions 2 and 3 depend on the schedule, so designs differ in their estimand for the same drug: the RCT's week-20 contrast is 1.5 times the Hybrid's β_D^{exp}. That difference is a real property of the designs (short re-exposures of a titrated drug), and it will appear in power. It means that power can be compared across designs at a common m_{BR}, but bias and coverage only against each design's own estimand value.

Implications under expectancy. For definition 2, the limits of `docs/40`, Section 9, at $t_{1/2} = 0$ are linear time -8.21, linear time with η_t -7.28, spline -6.22, visit factor -6.25, all under working independence and the published allocation. Against $\beta_D^{\text{exp}} = -7.22$ linear time overstates the effect by about 1 point and the expectancy model is close (inferred from those limits and the value above; the weightings differ slightly, so this must be recomputed in the closed-form step).

The test is common to all definitions. Under $m_{BR} = 0$ every definition is zero, so size and power refer to the same null hypothesis in every design. The estimand choice affects bias, coverage and the interpretation of $\hat{\beta}_D$, not the rejection rates.

4.2 Recommendation (open decision)

We recommend **definition 2**, β_D^{exp}, as the primary estimand for estimation, with definition 3's randomized version reported as a secondary estimand for the Hybrid design. Reasons: it is defined for every design including the RCT; it is a pharmacological quantity, so expectancy confounding shows as bias rather than being built into the target; and it is attainable by analyses a trial can run. Its cost is that it is design-specific. This choice is open (Section 8, item 1). Its value in every cell is computed from the population means of the cell, in closed form, and stored with the cell.

4.3 Secondary estimands

- τ^2**, on the scale of the primary estimand.** Participant i's exposure-averaged effect is $-(m_{BR} + v_i)\,\bar{q}_d$, with $\bar{q}_d$ the design's on-minus-off mean of q_{it}; its variance is $\tau_d^2 = \tau^2 \bar{q}_d^2$ (derived). The random-slope analyses of Section 5 target τ_d^2, not τ^2; both are reported.
- **Individual effects.** The participant-level value $-(m_{BR} + v_i)\,\bar{q}_{d,p(i)}$, conditional on v_i and excluding the transient BR noise.
- **Size.** The rejection rate under $m_{BR} = 0$, with and without heterogeneity.

5. Methods (M)

5.1 Closed form first

Before any simulation, every analysis whose mean model is fixed is evaluated in closed form, extending `docs/37` and `docs/39` from the biomarker slope to the drug coefficient.

- **Probability limit.** For an analysis with mean design X_p on path p and working covariance V, $\operatorname{plim} \hat{\beta}_D = \left[\left(\sum_p \varpi_p X_p' V^{-1} X_p\right)^{-1} \sum_p \varpi_p X_p' V^{-1} \mu_p\right]_D$,

with $\varpi_p = 1/P$ the expected path share (local symbol; π is reserved for power) and μ_p the true mean vector (`compendium-mathematics.tex`, "Time adjustment and the probability limit of the drug coefficient", extended from working independence to a general V; derived). Its difference from the estimand value is the bias, and under $m_{BR} = 0$ a nonzero limit predicts size inflation. Linear time without η_t is predicted to have a nonzero limit under the null in the Hybrid and OL+BDC designs (inferred from the 2-point difference between linear time and visit effects in `docs/40`, Section 9).

- **Variance.** The sandwich $\mathrm{Var}(\hat{\beta}_D) = N^{-1}[B^{-1}(\sum_p \varpi_p X_p' V^{-1} \Sigma_p V^{-1} X_p) B^{-1}]_{DD}$, $B = \sum_p \varpi_p X_p' V^{-1} X_p$, with Σ_p the true covariance of $\mathbf{Y}_i$: the refined separable covariance plus BL variance, plus $\tau^2 \mathbf{q}_p \mathbf{q}_p'$, plus measurement error. For the summary-measure analyses PW and PR this is exact (the contrast is a fixed linear combination, as E9 is in `docs/37`, Section 4); for the likelihood analyses it holds at the probability limit of the working covariance parameters, which must be computed by fitting the working model to Σ_p (approximate).
- **Contrast variance by design.** `docs/39`, Section 3, gives the contrast variances of the on-minus-off weights by design under CS and AR(1) (Hybrid 107.3 to 126.8 under AR(1) at $\rho = 0.8$ without measurement error, against 212.8 for CO; to be recomputed at the refined $\rho = 0.7$). For the main effect the same quadratic form $a'Ca$ applies, with two changes: the drug signal is $a'\mu_{BR}$ instead of the coupling term, and the weights must also remove the non-drug mean, either by orthogonality to the time and expectancy terms or by the between-path construction. The heterogeneity adds $\tau^2(a'q)^2$.
- **RCT.** For ANCOVA at week 20 with r_{BL}, $\mathrm{Var}(\hat{\beta}_D) \approx (2/N)(2\sigma^2_{20|Y_0^*} + \tau^2 q^2_{20})$ at equal arms, with $\sigma^2_{20|Y_0^*} = 598.5 - 18.48^2\, r_{BL}$ the placebo-arm residual variance (derived). At $N = 70$ and $\tau = 0$: SE 3.83 at $r_{BL} = 1$, 4.08 at 0.9, 4.31 at 0.8; normal-approximation power at the published ceiling 0.82, 0.77, 0.72 (verified, arithmetic; ignores the t reference distribution).
- **Analytic power curves.** Power by noncentral t at the analysis's reference degrees of freedom, over the f_m grid, for every design. The simulation then validates the closed form in the cells of block 1 and of blocks 2 and 4: agreement within 2 MCSE for rejection rates and within 2 MCSE for the empirical SD, as `docs/37`, Section 6 and `docs/39`, Section 9 required. Disagreement is reported, not tuned away.

The closed form is also the gate for the grids: if it shows a design at power above 0.99 or below 0.10 across most of the grid, the grid is revised before the run, and the revision is recorded.

5.2 Analyses

The analyses adapt the fourteen stress-test analyses (`compendium-mathematics.tex`, Table `tab:invd-analyses`; `docs/47`) to the main effect by dropping the biomarker terms, and add analyses that only the main effect needs. IDs shared with the stress test denote the main-effect analog of the same model. Notation: X_{it} is $D_{bc,it}$ at $t^a_{1/2}$, with the pre-exposure guard (0 at baseline and before first exposure; paper 01, Appendix B.4), and $X_{it} = D_{it}$ when $t^a_{1/2} = 0$. Default $t^a_{1/2} = 0.2$ week. All N-of-1 analyses use the long data with the observed baseline as a row at $t = 0$ unless stated. RC is `nlme::lme` with a random intercept and `corCAR1(form = ~ t | ptID)` on weeks, REML; RCH adds `varIdent(form = ~ 1 | phase)` (baseline, open label, blinded); CR2 is `clubSandwich::coef_test(vcov = 'CR2', test = 'Satterthwaite')` clustered on participant [5, 6].

Table 7: Analyses of the drug main effect: mean model, covariance, inference, degrees of freedom and purpose

ID	Mean model	Covariance	Inference	df	Purpose
S	$t + X$	RC	CR2	Satterthwaite (CR2)	stress-test reference; exposes the expectancy confounding
F1	$t + X + \eta_t$	RC	CR2	Satterthwaite (CR2)	expectancy-term model; proposed reference (strawman)
G1	as F1	RCH	CR2	Satterthwaite (CR2)	phase variances
G2	as F1	RCH	model-based	containment	phase variances, model-based
NS	`ns(t, 3)` $+X + \eta_t$	RC	CR2	Satterthwaite (CR2)	flexible time (`docs/42`)

ID	Mean model	Covariance	Inference	df	Purpose
VF	visit factor $+X$	RC	CR2	Satterthwaite (CR2)	randomized between-path contrasts only
A	$t + X$	unstructured, `mmrm`	Kenward-Roger	Kenward-Roger	stress-test analog; known liberal at small N
VA	visit factor $+X$	unstructured, `mmrm`	Kenward-Roger	Kenward-Roger	the MMRM convention [7]
R	$t + D$	compound symmetry, `mmrm`	model-based	Satterthwaite (`mmrm`)	published-type reference [4]
XS	$t + D + \eta_t$, first off-drug visit after each discontinuation dropped by schedule	RC	CR2	Satterthwaite (CR2)	period exclusion without oracle information; versus D_{bc}
XO	as XS, dropping visits whose true carryover fraction exceeds 0.05	RC	CR2	Satterthwaite (CR2)	oracle; bridge to the v1 analysis
PW	per participant, blinded on-minus-off mean; one-sample t	none	exact t	$N - 1$	E9 analog; time-confounded by construction

ID	Mean model	Covariance	Inference	df	Purpose
PR	per participant, $s_i = \sum_t (D_{p(i)t} - \bar{D}_t) Y_{it}$; path-stratified mean	none	t	$N - P$	randomization-based summary measure (the FWL estimator)
RS	as F1, random slope on X correlated with the intercept	RC + slope	CR2	Satterthwaite (CR2)	τ^2, individual effects
RSL	as RS without η_t	RC + slope	CR2	Satterthwaite (CR2)	τ^2 under linear time
OP	$t + \eta_t + Q$, $Q_{it} = q_{it}$ the true profile	RC	CR2	Satterthwaite (CR2)	oracle shape; targets β_D^{full}
AN	RCT: week-20 score on arm and Y_{i0}^*	OLS	HC2	$N - 3$	RCT primary
AN0	as AN on the error-free BL_i	OLS	HC2	$N - 3$	oracle; reproduces the v1 RCT
AM	RCT: all visits, visit-by-arm, Y_{i0}^* covariate, contrast at week 20	unstructured, `mmrm`	Kenward-Roger	Kenward-Roger	RCT using every visit

Notes on the table:

- **Kenward-Roger** is the `mmrm` implementation. The stress test used its linear variant (`Kenward-Roger-Linear`; compendium, "Inference"); the same variant is used here for comparability, which should be confirmed (Section 8, item 4) [8].

- **CR2 as default.** CR2 with Satterthwaite degrees of freedom is the default inference for every random-intercept model, because model- based and Kenward-Roger inference were anticonservative with flexible time in the Hybrid carryover cell and CR2 was not (`docs/42`, update of 2026-10-07; `docs/45`, Section 4.4), and A failed the size screen of the stress test (0.087 at $N = 35$; `docs/47`, Section 3).
- **HC2** for the RCT is the singleton-cluster case of CR2; it protects against the unequal arm variances that $\tau > 0$ creates.
- **F1 in CO.** With η_t constant after baseline, η_t is a baseline indicator there, so F1 frees the baseline mean.
- **XS against D_{bc}.** XS replaces the v1 oracle exclusion by a rule a trial can apply. In the Hybrid design it drops weeks 11 (paths 1 and 2) or 10 (paths 3 and 4) and the first post-crossover off-drug visit.
- **PW and PR** are the main-effect counterparts of E9. PW keeps the time confounding of a within-path contrast (null bias +0.37 points over blinded visits; Section 4.1) and is included because its moments are exact and it shows the size of the confounding. PR uses only the visits at which paths differ in drug state, so its mean is the randomized contrast and its variance is exact.
- **Random slopes.** In the Hybrid design each participant has four or five on-drug and three or four off-drug visits, so the slope variance may be weakly identified and hit the boundary. τ_d^2 is tested by the likelihood ratio against RC with the 50:50 mixture of χ_1^2 and χ_2^2 [9]; its interval is the `nlme::intervals()` approximation (Wald on the log SD scale), whose coverage is itself a result. Boundary and nonconvergence rates are reported.
- **Misspecified half-life.** S, F1, RS and XS are refitted at $t_{1/2}^a \in \{0, 0.5, 1\}$ in the cells named in Section 3.6. The AIC-selected half-life is not included: it failed the size screen with the expectancy terms in the stress test (F1A, 0.057 pooled; `docs/47`, Section 8).

6. Performance measures (P)

6.1 Measures

For each analysis and cell, with MCSEs as in Morris, White and Crowther [1], Table 6:

Table 8: Performance measures and their Monte Carlo standard errors

Measure	Definition	MCSE
Size (null cells)	$\hat{\pi}_0 = n_{\text{reps}}^{-1} \sum I(p_r < 0.05)$	$\sqrt{\hat{\pi}_0(1-\hat{\pi}_0)/n_{\text{reps}}}$

Measure	Definition	MCSE
Power	as above, alternative cells	as above
Size-adjusted power	rejection at the 5% quantile of the analysis's null p-values in the matching null cell	binomial, conditional on the quantile
Bias	$\bar{\beta} - \beta_D$, against the cell's estimand value	$\mathrm{SD}(\hat{\beta})/\sqrt{n_{\mathrm{reps}}}$
Empirical SE	$\mathrm{SD}(\hat{\beta})$	$\mathrm{SD}(\hat{\beta})/\sqrt{2(n_{\mathrm{reps}}-1)}$
Model SE calibration	$\kappa = \overline{\mathrm{SE}}/\mathrm{SD}(\hat{\beta})$	delta method [1]
Coverage	share of nominal 95% intervals containing β_D	binomial
τ_d^2 bias, coverage	as above for the RS and RSL estimates and intervals	as above
Individual effects	RMSE and correlation of the BLUP with the true individual value	bootstrap over replicates
Failed fits, boundary fits	rates	binomial

Failed fits are excluded from rates and counted, as in the stress test (`docs/46`, Amendment 2, item 3); an analysis with more than 2% failed fits in any cell is flagged.

6.2 The nominal band

At R1, a single null cell has MCSE 0.0049 and a 2-MCSE band of 0.040 to 0.060; the pooled rate over the 51 N-of-1 null cells of one analysis across designs has MCSE about 0.0007. At R2 the single-cell band is 0.036 to 0.064, as in `docs/46`.

6.3 Paired comparisons

All analyses of a design are fitted to the same trials, and power differences within a cell are tested by McNemar's test. Designs are not paired: their schedules differ. Participant-level draws that do not depend on the schedule (BL_i, v_i, ε_{i0}) use common random numbers across designs; design comparisons are otherwise treated as independent proportions.

6.4 Working decision rule

Within each design, adapted from `docs/46`, Section 7, and stated as a working rule that may be revised with recorded reasons:

1. **Size screen.** Discard an analysis whose pooled null rejection rate over that design's null cells lies outside 2 MCSE of 0.05, or that exceeds 0.075 in any null cell. Heterogeneous nulls (block 2) count; they test whether ignoring τ^2 costs size.
2. **Estimation screen** (new). In the base alternative cells, flag an analysis whose coverage of β_D is below 0.925 or whose absolute bias exceeds 10% of $|\beta_D|$. Flagged analyses may still be reported for testing, not for estimation.
3. **Oracle exclusion.** XO, OP and AN0 are never selected.
4. **Selection.** Among survivors, select by minimax regret: the smallest largest shortfall in size-adjusted power over the alternative cells.
5. **Margin.** The proposed reference F1 is retained unless a survivor beats it by more than 0.05 in largest shortfall, with the McNemar difference significant at 0.05 in at least half the alternative cells. Ties within 0.02 go to the analysis with fewer covariance parameters.

Across designs. Power curves of each design's selected analysis are compared at common m_{BR} and N; differences are tested as independent proportions. No design is "selected": the comparison reports operating characteristics.

7. Implementation plan

7.1 What to reuse

- **The stress-test driver** `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stres` (inspected): its `build()` already gives anchored carryover and randomized allocation by default (lines 25 to 35, 118 to 133), the separable AR(1) construct, Weibull decay, dropout, measurement error, per-chunk seeds and checkpointing. It is the basis of the new driver.
- **The patches** `analysis/patches/orig-to-refined/` implement the refined DGP and fifteen analyses selectable by `op$analysis`, verified against the driver. They are the reference for reproducing the refined process in Hendrickson's code, but not the engine: their `generateSimulatedResults()` fits the analysis once to the whole population, which its README reports as impractical with the refined analyses (a single F1 fit on 7,200 participants did not finish in 10 minutes), and their analyses test `bmc:Dbc` only.
- **Design definitions** from `hendrickson-problems/04-power-simulation.R` (lines 74 to 97).

7.2 What to add

1. **Closed-form module** (`01-closed-form.R`): estimand values per cell; probability limits and sandwich variances for every fixed- mean analysis; analytic power curves. Runs first; gates the grid.
2. **Drug-effect scale.** The BR, TV and PB ceilings are hard-coded in `build()` (10.98604 at line 120; 6.50647 at lines 134 and 135); make them cell parameters.
3. **Random drug slope.** $v_i\, q_{it}$ added after the draw, with q_{it} from the unit-ceiling profile.
4. **Designs.** OL+BDC and CO in the driver (11 uses Hybrid only, line 72); the RCT arm; the optional multi-crossover.
5. **Observed baseline with error** for all designs.
6. **Main-effect analyses** of Section 5.2, each returning the estimate, SE, df, p, interval, and for RS and RSL the variance component, its test and the BLUPs.
7. **Unit checks** before the run: the estimand values reproduce Section 4.1 at the published ceiling; the RCT closed form matches a large draw; τ^2 is recovered from the generated individual effects; allocation shares; the closed form against a direct Monte Carlo of PW and PR.

7.3 Compute estimate (inferred)

From `RERUN-anchored.md`, Section 3: the stress test costs about 11 worker-seconds per trial for about twelve analyses, dominated by three unstructured Kenward-Roger fits; its post hoc runs cost about 0.4 worker-seconds per random-intercept fit with CR2. The list above has nine random-intercept fits, two unstructured fits, one `mmrm` compound symmetry fit, two random-slope fits and two summary measures, about 11 worker-seconds per N-of-1 trial if a random-slope fit costs 1 second (assumed). At R1 that is about 165,000 N-of-1 trials, 500 worker-hours: about 63 hours on 8 cores, 170 hours at simrun's default of 3 cores. R2 halves this. The $N = 150$ cells add perhaps 10% (assumed). RCT fits are negligible. A pilot of 20 replicates per cell is run first to time the fits, as in `docs/46`, Section 9.

7.4 Execution on the simulation server

Through `simrun`, as described in `RERUN-anchored.md`, Section 4.2: the job runs the committed tree, so the driver must be committed first, and `simrun submit` refuses while tracked files have uncommitted changes; commands are argument vectors without redirection; the job's `SIM_CORES` sets the worker count; results come back with `simrun fetch` and `simrun apply`. One job per block keeps jobs short and restartable. The core count the host can give is unknown (unverified).

7.5 Output naming

- Drivers: `analysis/scripts/treatment-main-effect/rebuild/`, `01-closed-form.R`, `02-simulate.R`, `03-decision.R`, `04-tables-figures.R`. The v1 drivers stay where they are.
- Data: `analysis/data/treatment-main-effect/rebuild/`: `closed-form-{estimands,plim,power}.c` `cells-{tag}/` with chunk files; `replicates-{tag}.rds`; `summary-{tag}.csv`; `decision-{screen,estimation,power,minimax}-{tag}.csv`. Tags `full`, `pilot20`, and a block suffix (`-b1` to `-b9`) for server jobs.
- Seeds per chunk, $20261009 + 1000 \times \text{cell} + \text{chunk}$, so results do not depend on the number of cores (the convention of `docs/47`, Section 11).

8. Open decisions for the user

1. **Estimand definition.** Adopt β_D^{exp} (exposure- averaged on-minus-off drug response at the design's own visits) as primary, with the randomized blinded contrast as secondary for the Hybrid design, or choose the full-effect or the blinded contrast as primary (Section 4).
2. τ^2 **grid.** $\tau \in \{0, 2.75, 5.49\}$ points (0, 0.25 and 0.5 of the published ceiling), set in absolute units so that heterogeneous nulls exist; or a grid relative to m_{BR}.
3. **Effect-size grid units.** The multiplier f_m on the drug-response ceiling, reported in points and in units of $\sigma_{Y|\text{BL}} = 16.03$; or another SD (the unconditional 24.46, or an RCT-based Cohen's d).
4. **Analysis list.** The nineteen analyses of Section 5.2, including the oracle arms XO, OP and AN0, the variant of Kenward-Roger, and the default $t_{1/2}^a = 0.2$ week; drop or add analyses.
5. **Multi-crossover.** Whether to keep the current paper's design as an additional design (bridge to v1, expected to have little power under retitration), and if so in which blocks.
6. **Replicate counts.** R1 (2,000 null, 1,000 alternative; about 63 hours on 8 cores) or R2 (1,000 and 500, as `docs/46`; about 32 hours).
7. **RCT baseline.** Whether the RCT is analyzed by ANCOVA on a baseline measured with error, and how much error: reliability 1 (error-free, as v1), 0.9 or 0.8; and whether the same observed baseline is used in the N-of-1 designs (proposed: yes).

Further decisions that the protocol has had to assume:

8. **Base half-life** 0.2 week, with 0.1 optional and 1 week as the labeled behavioral scenario.
9. **Form of heterogeneity.** v_i scaling the Gompertz profile (proposed), or scaling the binary drug state; independent of BL_i (proposed) or correlated with it.

10. *BR* **noise in unexposed participants.** Keep the inherited SD 8 off drug and in the RCT placebo arm (the refined process as specified), or make the *BR* SD follow the drug effect.
11. **Decision rule.** The estimation screen of Section 6.4, item 2, is new relative to `docs/46`; its thresholds (coverage 0.925, bias 10%) and the single-cell cap of 0.075 at R1 (5 MCSE) need confirming.

From the literature review (Appendix A). Proposals only; none is applied to Sections 1 to 7. Reasons and sources are in Appendix A.5.

12. **Randomization test (L1).** Add a permutation test that re-draws the path labels under the trial's own allocation rule, applied to the PR statistic in every cell and to F1 in a subset of cells.
13. **Bayesian hierarchical arm (L2).** Add a Bayesian random-slope analysis in blocks 1 to 3 at $N = 35$ and 70, judged by the same frequentist operating characteristics, or record its omission as a stated limitation.
14. **Onset exclusion by schedule (L3).** Add an analysis XT that drops the first on-drug visit after each start or restart of the drug, the onset counterpart of XS.
15. **Literature bridge cell (L4).** Add a reduced cell (constant effect, no PB or TV, $t_{1/2} = 0$, AR(1) only) in which PW and the random-slope model can be checked against the published closed forms of Senn [2], Araujo et al. [11] and Yang et al. [16].
16. **Heterogeneous null with carryover (L5).** Cross block 2's null cells with the behavioral half-life $t_{1/2} = 1$ week.
17. **Two-point heterogeneity (L6).** Add one cell in which v_i takes two values (responders and non-responders) with the same τ^2 as a normal cell; this extends item 9.
18. **RCT comparator for the design comparison (L7).** Report AM, which uses every visit, as the information-matched comparator in the across-design comparison, and AN as the conventional comparator.
19. **Additional measures (L8).** Add interval width for β_D, coverage of individual-effect intervals, and a per-participant OLS comparator for the BLUPs.
20. **Bounded outcome (L9).** Add one sensitivity cell in which Y is rounded and truncated to the CAPS range.

9. What carries over from the current paper

Carried over, with edits (`report.Rmd`):

- The Introduction's background on N-of-1 designs and sample size (lines 76 to 141), including the Senn and Zucker variance-component apparatus (lines 142 to 252), which becomes directly relevant once τ^2 is in the DGP; references [2, 3, 10, 11] stay.

- The single-case framework, prazosin, regulatory and literature sections (lines 290 to 458), with the comparisons of Blackston et al. [12] and others, rewritten where they quote the paper's own results.
- The comparison with prior simulation studies in the Discussion (lines 1770 to 1922), kept as structure, with every number replaced.

Replaced. The abstract; the Objectives (line 460); all of Methods (lines 472 to 600); all Results (lines 1053 to 1573); the remaining Discussion and the Conclusions; the cover letter, title page, short report and bullets. The word "hybrid" is removed from every statement about the v1 simulation.

Bibliography fix found while reading. The key `araujoComparisonFourMethods2014` in the paper's `references.bib` (line 22) holds Chen and Chen (2014); the key should be renamed (inspected).

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References 16 to 61 were added on 2026-10-08 with Appendix A. Each was checked against at least one of a PubMed record, Crossref, an arXiv abstract

page or the Zotero record; the source checked and the extent of reading (full text or abstract only) are listed in Appendix A.1, Table "Studies screened in full and how each was verified". Reference 16 is a preprint for which no journal version was found; reference 60 was checked against Crossref only and not read.

Changes to the working protocol

None recorded as of 2026-10-08. Changes made after results are seen will be appended as numbered amendments, each with its date, scope and reason.

Appendix A. Prior simulation and analytic approaches to testing the treatment main effect

This appendix describes how earlier work has evaluated tests and estimators of an average treatment effect in N-of-1, aggregated N-of-1, multi-crossover and single-case designs, and compares that work with Sections 3 to 6. It changes nothing in Sections 1 to 10; its proposals are collected in A.5 and entered in Section 8 as items 12 to 20.

Notation in this appendix. Each source's model is restated in the protocol's symbols where that is possible: Y_{it} the outcome, β_D the population mean treatment effect, $v_i \sim N(0, \tau^2)$ the participant-by-treatment deviation, ρ an AR(1) coefficient, $t_{1/2}$ a carryover half-life and N the number of participants. Two symbols are local to the appendix: σ_w^2, the within-participant residual variance of one occasion, and k, the number of AB cycles per participant; m is the number of observations in a single-person series. A source's own symbol is given once, in parentheses, when it differs. Several sources use τ or τ^2 for something other than treatment heterogeneity (a treatment effect, or a random-intercept variance); this is pointed out where it matters, because it is the most common source of misreading in this literature.

Evidence labels. As in the protocol, with one addition. *Quoted*: the source's words. *Inspected*: read in the source on 2026-10-08. *Derived*: follows algebraically from the source's model. *Argued*: a judgment of this team about relevance to the Hybrid setting, not a finding of the source. Unless a statement is labeled otherwise, statements about a source's design, model and results are inspected, from the reading named for that source in Table "Studies screened in full and how each was verified".

A.1 Search

Sources and dates. All searches were run on 2026-10-08.

- **Zotero, collection `pmsimstats-ng`** (key L29UF4HU): listed in full, 391 entries in two pages (200 and 191), including duplicate records of the same paper. The library has no collection named `nof1`; a library-wide keyword search for "N-of-1" returned 84 records, all but a handful already in the collection. The library's semantic search failed (an embedding-service authorization error) and was not used.
- **PubMed**, eight queries: N-of-1 (title) with simulation or Monte Carlo and power or type I error; series, aggregated, combined or meta-analysis of N-of-1 with simulation; single-case with multilevel or hierarchical and simulation; Ferron, Moeyaert or Van den Noortgate with single-case or multiple-baseline and simulation; N-of-1 or multiple crossover with sample size or power and autocorrelation, serial correlation or carryover; N-of-1 with Bayesian hierarchical and simulation or operating characteris-

tics; Zucker, Nikles, Stunnenberg or Schmid with N-of-1; multiple, multi-period, replicate or repeated crossover with simulation. The queries returned 119 records, with overlap.
- **Web search** for journals outside PubMed (statistics and psychometrics journals, arXiv, Harvard Data Science Review), three queries, and the reference lists of every paper read in full.
- **Repository sources**: paper 04's literature sections (`report.Rmd`, lines 142 to 458 and 1770 to 1922) and `references.bib`; `docs/51` (Section 2 and its reference list); `docs/references/nof1-simulation.bib`; `docs/29`.

Inclusion. A study was included in A.2 and A.3 if it evaluated, by Monte Carlo simulation or in closed form, a test or estimator of an average (or individual) treatment effect in an N-of-1, aggregated N-of-1, multi-crossover, AB/BA crossover with repeated measures, or single-case design, and reported at least one of size, power, bias, coverage or precision. Papers that only applied a method to one data set, tutorials, protocols with a single sample-size calculation, reviews and studies of omics "N-of-1" signals were excluded from the comparison and, where useful, are listed as background in A.2.10. Hendrickson et al. [4], whose target is the biomarker interaction, is described because its data-generating process is the ancestor of the refined process.

Counts. About 520 records were screened on title or abstract (391 Zotero entries, 119 PubMed records, and the web and reference-list leads, with heavy overlap between sources). Fifty-one were assessed in full or in abstract: 38 were included in A.2 and A.3, and 13 are kept as background (A.2.10). By where they are held, 26 of the 38 included studies are in the `pmsimstats-ng` collection; 11 were found through PubMed and 1 through the repository's `docs/references/nof1-simulation.bib`, and these 12 are not in the collection. Further leads from reference lists (for example on randomization tests for reversal designs, gradual-effect models and one-stage IPD meta-analysis) were identified but not assessed; they are listed in the report that accompanies this revision, not here.

How each item was verified. Reading was done by six parallel extraction passes in this session, each told to verify every record, read full text where it was available, and say where only an abstract was read. Full text from PubMed Central loses inline mathematics; where an equation was reconstructed from surrounding text or a publisher page, the extraction said so, and such equations are given below in words or flagged.

Table 9: Studies screened in full and how each was verified

Ref	Study	Record checked	Reading
[2]	Senn 2019	PubMed, Zotero	full text (Zotero PDF)

Ref	Study	Record checked	Reading
[11]	Araujo, Julious, Senn 2016	PubMed	full text (PLOS, PMC)
[16]	Yang, Steingrimsson, Schmid 2021	arXiv	full text (Zotero PDF, v1)
[17]	Hedges, Shadish, Natesan Batley 2023	Crossref, Zotero	full text (Zotero PDF)
[18]	Hedges, Pustejovsky, Shadish 2012	PubMed	full text (ERIC copy)
[19]	Shi, Ye 2024	PubMed	full text (arXiv v2); supplement not read
[12]	Blackston et al. 2019	PubMed	full text (PMC)
[20]	Carrozzo et al. 2025	PubMed	full text (PMC)
[21]	Chen, Chen 2014	PubMed	full text (PMC, PLOS); 4-cycle supplement not read
[22]	Gärtner et al. 2023	PubMed	full text (PMC); supplement not read
[23]	Krone et al. 2020	PubMed	full text (PMC)
[24]	Tang, Landes 2020	PubMed	full text (PMC); figures not read
[25]	Wang, Schork 2019	PubMed	full text (PMC); power figures not read
[26]	Lin et al. 2016	PubMed	full text (PMC); appendix tables not read
[27]	Percha et al. 2019	PubMed	full text (PMC); equations partly reconstructed
[28]	Schork 2022	Crossref, Zotero	full text (Zotero PDF, author manuscript)
[29]	Liao et al. 2023	PubMed	full text (PMC)
[30]	Selukar, Prince, May 2025	PubMed	full text (PMC); supplement not read

Ref	Study	Record checked	Reading
[31]	Jiang, Arnold, Betensky 2025	PubMed	full text (PMC)
[32]	Senarathne, Overstall, McGree 2020	Crossref, Zotero	full text (arXiv v3)
[33]	Shrestha, Jain 2021	PubMed, Zotero	**abstract only**
[34]	Giai et al. 2023	PubMed, Crossref, Zotero	**abstract only**
[35]	Cruz, Melo, Mylona 2026	Crossref, arXiv	full text (arXiv v3, not the published version)
[36]	Ferron et al. 2009	PubMed	full text (publisher PDF)
[37]	Ferron, Farmer, Owens 2010	PubMed	full text (publisher PDF)
[38]	Moeyaert et al. 2013 (BRM)	PubMed	full text (publisher PDF)
[39]	Petit-Bois et al. 2016	PubMed	full text (publisher PDF)
[40]	Declercq et al. 2019	PubMed	full text (publisher PDF)
[41]	Declercq et al. 2022	PubMed	full text (ERIC copy)
[42]	Moeyaert, Yang, Xu 2022	PubMed, Crossref, Zotero	PMC page through a summarizing reader only
[43]	Chen, Pustejovsky 2024	PubMed	full text (accepted manuscript, PsyArXiv)
[44]	Bouwmeester, Jongerling 2020	PubMed	full text (PMC)
[45]	Michiels et al. 2020	PubMed	full text (author manuscript, KU Leuven)
[46]	Levin, Ferron, Kratochwill 2012	PubMed, Crossref, Zotero	**abstract only**

Ref	Study	Record checked	Reading
[47]	Tueller et al. 2023	PubMed	full text (PMC); table values through a summarizing reader
[48]	Natesan Batley, Hedges 2021	PubMed	full text (PMC)
[49]	Joo, Ferron 2019	PubMed	**abstract only**
[4]	Hendrickson et al. 2020	PubMed	full text (local PDF)

The background items of A.2.10 were checked as follows: [3], [50], [51], [53], [54], [56], [57] and [55] against PubMed and read in full (for [53], the Zotero full-text cache); [52] against PubMed, abstract and targeted passages of the PMC page; [58] and [59] against PubMed, abstract only; [60] against Crossref only, not read (no abstract in Zotero, publisher page not accessible); [61] against PubMed, not read.

A.2 The studies

The studies are grouped by what they contribute. Within each, the extraction terms of the task are given in a fixed order: design; DGP (effect form, onset, carryover, serial correlation, heterogeneity, expectancy and trend, missingness, outcome); estimand; analyses; inference; performance measures and MCSE; scale; comparator; conclusions. "nr" means not reported in what was read.

A.2.1 Closed-form variance-component frameworks Senn 2019 [2]. Closed form, no simulation. Design: N participants each randomized within $k \geq 2$ cycles of a pair of occasions. Model on within-cycle differences: $d_{ij} = \beta_D + v_i + e_{ij}$, $v_i \sim N(0, \tau^2)$ (Senn's treatment-by-patient interaction), $e_{ij} \sim N(0, 2\sigma_w^2)$; all terms independent, no serial correlation, trend or period effect; carryover is assumed removed by a washout or by measuring late in each occasion (quoted: "carry-over can be eliminated either by using a washout or by limiting measurement to the latter part of any treatment occasion"). Estimands: β_D, individual effects (naive and shrunk), and the variance components. Analyses: the summary-measures random-effects analysis, $\mathrm{Var}(\widehat{\beta}_D) = (\tau^2 + 2\sigma_w^2/k)/N$ on $N-1$ degrees of freedom; the fixed-effects analysis for the participants studied, $2\sigma_w^2/(Nk)$ on $N(k-1)$ degrees of freedom, with the remark that software assuming $N-1$ misstates power; shrinkage standard errors that treat the weights as known. Worked example: with $\tau^2 = 1$, $\sigma_w^2 = 4$, $\beta_D = 1$, $k = 3$, the random-effects analysis needs $N = 31$ and the fixed-effects analysis $N = 22$ for

80% power. Conclusion (quoted): "very different purposes for n-of-1 trials may require very different samples sizes."

Araujo, Julious and Senn 2016 [11]. Mainly theory, with a small simulation. Model: $Y_{irs} = \lambda_i + \beta_{ir} + Z_{irs}(\beta_D + v_i) + \varepsilon_{irs}$, $Z = \pm 1/2$, with a patient effect, a cycle-within-patient effect, $v_i \sim N(0, \tau^2)$ (their ψ^2) and independent error; no carryover, serial correlation, onset or missingness in the simulation. The central closed-form result (derived in the paper): the true variance of the mean effect is $\tau^2/N + 2\sigma_w^2/(Nk)$, while the matched-pairs t-test over all cycles estimates $\{(N-1)/(Nk-1)\}\tau^2/N + 2\sigma_w^2/(Nk)$, which is too small whenever $\tau^2 > 0$ and $k > 1$; under the strict null ($\tau^2 = 0$) the matched-pairs test is valid. Simulation: τ^2 from 0 to 1, $\sigma_w^2 = 1$, $k = 3$, 1,000 trials per value, comparing empirical and model variances (figures only; no size, power or MCSE). They recommend allowing for the random interaction when computing intervals, and (quoted) "We do not recommend attempting to adjust for carry-over but instead recommend using washout periods of adequate length." They also argue that AR(1) errors alone do not allow for a between-patient main effect and should be applied to within-patient errors only. The paper is in part a critique of Chen and Chen [21].

Yang, Steingrimsson and Schmid 2021 [16] (preprint; no journal version found). Closed form, no simulation. LMM $Y_{ijkl} = \mu_{ij} + \delta_{ij} A_{ijkl} + \varepsilon_{ijkl}$ with fixed or random intercepts crossed with common or random slopes (random slope variance τ^2, their σ_δ^2), and independent, exchangeable or AR(1) residuals, with several measurements per period. Standard error by generalized least squares with known variance components and normal-approximation power; a Shiny tool. Assumed absent (quoted): "no underlying trend in the outcome or carryover effects." Findings: random-slope models need more total measurements than common-slope models, and the residual structure changes the required sample size. The authors note that a t rather than normal reference would be needed in small samples.

Hedges, Shadish and Natesan Batley 2023 [17]; Hedges, Pustejovsky and Shadish 2012 [18]. Closed-form power ([17]) and the effect-size theory behind it ([18]) for multi-case $(AB)^k$ reversal designs. Model: $Y_{it} = \mu_{\text{phase}} + \eta_i + \varepsilon_{it}$ with a random intercept η_i of variance written τ^2 in these papers (an intercept variance, **not** treatment heterogeneity: the treatment effect is the same for every case) and stationary AR(1) errors running across phases. Effect: an immediate, constant mean shift; no onset, carryover, trend or expectancy. The test statistic is the average within-person contrast over a pooled between-case SD, referred to a noncentral t with a Box-approximation degrees of freedom that depends on ρ and on the intraclass correlation, both treated as known in [17]. In [17], power is non-monotone in ρ, with a minimum near 0.6 to 0.8; with $k = 2$ pairs, 3 observations per phase, $\rho = 0.5$, intraclass correlation 0.5 and a standardized effect of 0.75, 80% power needs 7 cases. [18] simulated the $(AB)^1$ case only (8,000 iterations per cell) and found the variance of the estimator underestimated when the nuisance parameters are estimated; it evaluates estimation, not test size.

Shi and Ye 2024 [19]. Closed form with simulation, AB/BA two-period crossover against a parallel-group design. Potential-outcomes model in which carryover is a fixed additive shift in period 2 (λ_0, λ_1), with a period (natural-course) change and, in one case, treatment-by-covariate interactions. Closed-form bias of the basic crossover estimator, $\frac{1}{2}(\theta_1 + \tilde{\theta}_2 - \lambda_0 - \lambda_1)$, and closed-form size and power. Conclusion (quoted): "when the carry-over effect is positive, the basic estimator underestimates the treatment effect, which does not inflate the type I error of one-sided tests but negatively impacts the power"; negative carryover inflates size. With within-person correlation 0.3, 0.5 and 0.7, the crossover beats a parallel trial only if the mean carryover is below 0.41, 0.50 and 0.61 times the alternative effect. Inference is asymptotic z with design-based covariate adjustment; 10,000 replicates per grid point in one case, 2,000 in a resampling study; no MCSE. "Behavioral" carryover in this paper is a learned change in behavior that persists, the sense in which the protocol labels its $t_{1/2} = 1$ week scenario.

A.2.2 Aggregated N-of-1 trials against parallel and crossover trials
Blackston et al. 2019 [12]. Design: aggregated N-of-1 with 3 cycles of one treatment and one placebo observation (6 per participant); the crossover comparator uses cycle 1 (2 observations), and the parallel comparator one observation; all three are cut from the same simulated data; N from 10 to 200. DGP: $Y_{ick} = \mu_i + \beta_D Z_{ick} + \varepsilon_{ick}$ with a random intercept μ_i of SD $\sigma_\mu \in \{0, 0.1, 0.5\}$ and error SD $\sigma_\varepsilon \in \{0.5, 1\}$, in four scenarios; $\beta_D = 0.25$ (written τ in the paper). The "heterogeneity" of the scenarios is intercept variance; the treatment effect is common to all participants except in a selection-bias mixture in which a fraction of participants comes from a subpopulation with a different effect. Carryover: a constant (0.05, 0.10 or 0.15) added to the next observation after any treatment observation, regardless of time (quoted: "we do not consider length of cycle or washout"), present also when $\beta_D = 0$. No serial correlation, trend, onset or expectancy. Analyses: `lme4::lmer` with a random intercept, treatment tested by a deviance (likelihood-ratio) test; ordinary regression for the parallel trial; an extra model with an indicator for treatment in the previous period. Inference asymptotic; REML or ML not stated. 5,000 replicates per configuration; MCSE nr. Results without carryover: the aggregated design reaches 80% power at $N = 30$ in three scenarios and $N = 100$ in the fourth; the parallel trial needs $N = 150$ and the crossover $N = 100$; size 0.05 for the aggregated design, 0.05 to 0.07 for the crossover. With carryover: size of the aggregated design 6% to 8% at 0.05, 14% to 15% at 0.10, 23% to 27% at 0.15; the crossover 6% to 17%. Under the selection mixture with half the sample from the other subpopulation, the aggregated design's size reached 37% to 40% in three scenarios. Conclusion (quoted): "Carryover effects can result in inflated type I error probabilities for both crossover and especially N-of-1 designs." Whether the previous-treatment adjustment removed the inflation is ambiguous in the text read.

Carrozzo et al. 2025 [20]. Design: parallel trial, AB/BA crossover, and an

aggregated N-of-1 design of 3 cycles in which each participant repeats one sequence; 10 to 100 per arm or sequence. DGP: $Y_{ijk} = \gamma_i + \beta_D Z_{ijk} + \varepsilon_{ijk}$, a random intercept γ_i with variance written $\tau^2 \in \{0, 0.01, 0.25\}$ (intercept variance, cancelling in within-person differences), and error SD drawn per participant from $U(0.05, 1)$. Carryover: 10% of β_D added to treatment-A observations that follow B; because it is proportional to β_D it vanishes under the null, so it cannot affect size by construction (derived). No serial correlation, trend, onset, expectancy or dropout. Analyses: unpaired t (parallel), period-difference t (crossover), and a two-stage random-effects meta-analysis of per-participant mean differences with a z interval; a sequential cumulative meta-analysis with a Bonferroni correction. 5,000 replicates (100 for the sequential analysis); MCSE nr. Results: the N-of-1 meta-analysis reaches 80% power at about $N = 30$ (40 with carryover), the crossover at 150, the parallel trial at 300 to 420; size about 0.05 for all. The paper states that the Kenward-Roger method was used as "the estimator of the between-study variance"; Kenward-Roger is an inference adjustment, not a variance estimator, so the method used is unclear.

A.2.3 Comparisons of analyses for aggregated trials Chen and Chen 2014 [21]. (Two authors; the citation "Chen, Chen and Xia" in the current paper 04 is wrong.) Design: aggregated N-of-1 with 3 or 4 AB cycles, one observation per period, $N \in \{1, 3, 5, 10, 20, 30\}$. DGP: multivariate normal with unit variances and compound symmetry (correlation 0, 0.5, 0.8) or AR(1) ($\rho =$ 0.5, 0.8) across periods; effects 2 for B and 2 to 3 for A (differences 0 to 1); carryover from the previous period only, 0% or 20% of the previous treatment's absolute effect, so placebo also carries over. No τ^2, onset, trend, expectancy or dropout. Analyses: M1, paired t over all pairs ignoring the participant; M2, a mixed model of within-cycle differences with a random participant effect; M3, a mixed model of the raw responses with period and carryover-indicator terms and a random intercept; M4, DerSimonian-Laird meta-analysis of per-participant mean differences. The fitted residual covariance of M2 and M3 was set to the generating one. Degrees-of-freedom method nr (SAS 9.2; the defaults were presumably used). 3,000 replicates; MCSE nr. Results (3 cycles): M1 and M3 near nominal size under compound symmetry; M2 severely conservative at small N (0.004 to 0.006 at $N = 3$); all methods conservative under AR(1) (M1 0.029, M3 0.015 to 0.021 at $N = 30$); M4 anticonservative at $N = 3$ (0.081 to 0.086) and biased upward by 26% to 182% of the true difference even without carryover, which suggests an implementation problem (argued). With 20% carryover, M1 and M2 were biased by about -10% and M3 was nearly unbiased. Conclusion (quoted): "We recommended paired t-test to be used for normally distributed data of N-of-1 trials because of its optimal statistical performance. In the presence of carryover effects, mixed effects model could be used as an alternative." Araujo et al. [11] show that this recommendation depends on $\tau^2 = 0$.

Gärtner et al. 2023 [22]. Design: aggregated N-of-1 comparing two active

treatments, daily measurements, two blocks of two 4-week periods; samples of 5 to 100 drawn from one generated population of 1,000. DGP: a directed acyclic graph of linear structural equations with time-varying covariates, a Wiener-process drift (natural course), constant effects (-2 and -4), and carryover through a recursive wash-in and wash-out indicator with time constants per treatment (the Percha form [27]); 10% missing rows with probability rising over time, plus a 10-day gap per person; an ordinal outcome analyzed as truncated Gaussian. No random effects or treatment heterogeneity (quoted: "we only included linear dependencies and fixed effects"), no expectancy, no explicit AR residual. Analyses: mean difference; OLS with covariates; a carryover-adjusted parametric model (COAPM) whose decay regressor's time constants are chosen by grid search; Bayesian networks; G-estimation via GEE with independence or AR(1) working correlation. The standard error reported is the SD of estimates over 100 subsamples, which are not independent replicates (argued); no tests, no coverage, no MCSE. Conclusion (quoted): "When there is carry-over, COAPM yields unbiased and more efficient estimates while all other methods show some bias"; the authors concede that COAPM was close to the generating process.

Krone et al. 2020 [23]. Estimation only. Aggregated N-of-1 with $N \in \{20, 30, 40\}$ and 10 to 30 measurements in random order; $Y_{it} = \beta_{0i} + (\beta_D + v_i)x_{it} + \varepsilon_{it}$ with a random intercept and random treatment effect (the text and a figure disagree on whether 0.5 is the variance or the SD of v_i); no serial correlation, carryover or trend. Two-stage meta-analysis against frequentist and Bayesian random-slope LMMs, judged by the RMSE of individual effects and the estimated heterogeneity; 1,000 replicates per scenario; no test of β_D. Conclusion: the mixed models are preferable, and the two-stage method overestimates heterogeneity.

A.2.4 Single-person series These studies analyze one participant. They bear on the protocol through the within-person part of the information and through the small-sample behavior of serial-correlation estimates.

Tang and Landes 2020 [24]. Single person, paired (multiple crossover) and two-sample designs, m from 4 to 12 and 30 to 100; $Y \sim N(X\beta, \sigma^2 V)$ with $V_{ij} = \rho^{|i-j|}$, $\rho \in \{-0.33, 0, 0.33, 0.67\}$; level or rate change; no carryover. Four "serial t-tests" using a bias-corrected lag-1 estimate of ρ and an effective sample size, against the usual t-tests; one-sided 5%; 10,000 replicates per cell with a stated Monte Carlo error target ("within 0.01 … with 95% confidence"). The serial tests improve on the usual tests, but (quoted) "Type I error is still substantial, power optimistic, and interval widths biased for small n"; they "do not account for carryover effects". Paper 04's specific values for this paper (size about 0.22 against 0.06 at $m = 12$, $\rho = 0.67$) come from figures that were not read here and are unverified.

Wang and Schork 2019 [25]. Single person, 400 observations in 2 to 80 alternating periods, fixed order, with or without washout; AR(1) errors; a constant effect of 0.3 SD; quoted: "we assumed a constant treatment response, not

a graded one … and no carry-over or intervention order effects for many of the analyses." Analytic power from the likelihood-ratio test under AR(1), checked by 1,000 simulations per setting fitting GLS with AR(1); asymptotic χ^2_1. Power falls with stronger serial correlation, which more frequent alternation partly mitigates; washouts reduce power slightly when there is no carryover. The current paper 04 attributes to this study a finding that washouts reduce effective sample size "in proportion to the carryover half-life"; that finding is not in the text read.

Lin et al. 2016 [26]. Single person, 28 daily observations, ABAB or AB; $Y_t = \alpha + \beta_D D_t + e_t$ with AR(1) e_t, $\rho \in \{0, 0.2, 0.5, 0.7\}$, normal or contaminated-normal errors. GLS with AR(1) by REML, tested by a Wald test with a normal reference, a parametric bootstrap and a residual bootstrap; 10,000 replicates per cell, with a stated tolerance interval for size (0.0457 to 0.0543). Wald size 0.061 to 0.113; parametric bootstrap 0.050 to 0.054; REML bias of $\hat{\rho}$ about -0.12 at $\rho = 0.7$.

Percha et al. 2019 [27]. Single person, blocks of randomly ordered periods. The most mechanistic onset-and-carryover DGP found: for each treatment, the effect state approaches its asymptote with a wash-in time constant while active and decays with a separate wash-out time constant when stopped (first-order kinetics); a Wiener-process baseline drift; mean reversion of the latent state toward drift plus effects; process noise and measurement error; discrete outcomes by link transforms. No heterogeneity, no expectancy component. Analysis: OLS with treatment and block effects, ignoring serial correlation, and a median-differencing rule; 50 trials per setting; no size, no MCSE. Treatment order, period length and number of blocks changed both power and the mean estimate (e.g. a true effect of 10 estimated at 5.8 to 12.0 across four orders). Recommendation (quoted): "our main recommendation for N-of-1 study designers is to simulate the study".

Schork 2022 [28]. Single person, 80 to 400 measurements in alternating blocks with unmeasured washout gaps; AR(1) with ρ from -0.99 to 0.99; no carryover. OLS t, `nlme::gls` (whether an AR(1) term was specified is unclear in the text), Newey-West HAC errors, and sequential probability ratio tests (standard and bootstrap) for single and aggregated series; 100 to 10,000 replicates. False positives rise with positive ρ and are worse with washout gaps; randomized order mitigates this; HAC errors help.

Liao et al. 2023 [29]. Single person, 120 daily observations with several switching schedules and no washout. A distributed-lag outcome model, $Y_t = \mu + \sum_{l=0}^{7} \beta_l X_{t-l} + \varepsilon_t$ with AR(p) errors, and five lag curves (geometric decay, slow absorption peaking at lag 2, non-monotone curves with sign changes, none). A Bayesian distributed-lag model with AR errors against three Bayesian lag models with independent errors; bias and RMSE of immediate, carryover and total effects; 100 replicates per scenario; no tests, no MCSE. Under-specifying the AR order hurt more than over-specifying it.

A.2.5 Sequential and adaptive designs **Selukar, Prince and May 2025 [30].** Single-person trials of 3 to 26 blocks of 2 to 6 periods, combined in series of 5 or 10 trials; normal data with a block random intercept; quoted assumptions "no carryover effect" and "no outcome drift"; one true effect for all trials of a series, so no heterogeneity (inferred). An LMM with a block random intercept, monitored by O'Brien-Fleming or Pocock boundaries, with naive, bias-adjusted and median-unbiased estimators; inverse-variance combination across trials. 10,000 trials per setting; MCSE nr. With 13 or fewer blocks, normal boundaries inflated size two to seven times, while (quoted) "all settings with t-adjusted boundaries have nominal or smaller than nominal type 1 error rates". Paper 04's summary ("reaches nominal rates with more blocks or alpha-spending corrections") should be revised to this.

Jiang, Arnold and Betensky 2025 [31]. Single-person cycles of two occasions with washout, monitored by O'Brien-Fleming boundaries on the cumulative mean difference with a three-way decision (superiority, equivalence, inferiority) and futility; individual effects from a three-point mixture $\{-\mu, 0, \mu\}$ with no within-class spread; a curtailed binomial rule across participants for the proportion of responders. Known-variance normal theory; 1,000 replicates; MCSE nr. Restricted by the authors to treatments with "rapid onsets and cessations of action". The population target is a proportion of individual classifications, not β_D.

Senarathne, Overstall and McGree 2020 [32]. Aggregated N-of-1 with 3 cycles; GLMM with random intercept and random treatment effect (variance 2.25 or 9, our τ^2), normal, log-normal and Poisson outcomes; no carryover or serial correlation. Compares Bayesian information-optimal adaptive allocation, a bandit allocation and fixed randomization by posterior precision and correct identification of each participant's better treatment; 20 repetitions per scenario; no frequentist size or power.

Shrestha and Jain 2021 [33] (abstract only). A Bayesian hierarchical model for individual and population effects with Thompson-sampling allocation in individual and aggregated N-of-1 trials, evaluated by a simulation motivated by a completed trial. DGP, replicates and measures could not be read.

Giai et al. 2023 [34] (abstract only). Individual and population Net Benefit from generalized pairwise comparisons with prioritized outcomes, in a simulation mimicking a 38-patient series; size was reported as good and power as low even at 140 days per patient. A nonparametric probabilistic estimand, not a mean difference.

A.2.6 Carryover in crossover designs with repeated measures **Cruz, Melo and Mylona 2026 [35]** (read as arXiv v3). AB/BA crossover with 10 to 100 measurements per period and 2 to 50 units per sequence; a within-period trend and sequence-specific carryover functions (sine and cosine) centered to mean zero; independent errors; no heterogeneity. GEE with exchangeable work-

ing correlation and sandwich errors, with and without penalized-spline carryover terms; 500 replicates; MCSE nr. All models estimated the treatment effect without bias, but models without carryover terms had coverage near 70%, and the proposed models near 95% as units or measurements increased. Because the carryover functions have mean zero, they cannot bias the treatment effect by construction (derived), so the study speaks to variance and coverage, not to bias.

A.2.7 Single-case designs: multilevel Monte Carlo studies All of these use multiple-baseline AB designs (none simulates ABAB reversals), an immediate level shift (sometimes with a slope change), no carryover, no expectancy and no dropout.

Ferron et al. 2009 [36]. $N \in \{4, 6, 8\}$, series of 10 to 30; $Y_{ij} = b_{0j} + b_{1j}\,\text{phase}_{ij} + r_{ij}$ with random intercept and random treatment effect (variance 0.1 or 0.3, our τ^2) and AR(1) level-1 errors, ρ from 0 to 0.4. REML LMM with independent or AR(1) errors, crossed with five degrees-of-freedom methods: containment, residual, between-within, Satterthwaite and Kenward-Roger. 5,000 replicates per condition; MCSE given once (0.0031 for coverage 0.95). With AR(1) modeled, Kenward-Roger coverage averaged 0.952 (range 0.935 to 0.965) and Satterthwaite 0.949; residual and between-within about 0.917; containment 0.972 with wider intervals. Without AR(1), coverage fell as ρ rose. The treatment- effect variance was overestimated (relative bias 0.34, 0.25, 0.21 at $N = 4$, 6, 8) and its interval covered about 0.83 to 0.85; the authors call these results "not encouraging".

Ferron, Farmer and Owens 2010 [37]. Same design and model; individual effects b_{1j} estimated by per-person OLS and by empirical Bayes from an AR(1) LMM; 1,000 replicates. Empirical Bayes estimates were shrunk (mean bias -0.24 on a true 1.0) but had lower MSE. Interval coverage: Kenward-Roger 0.958, Satterthwaite 0.920, containment 0.862, OLS 0.899. The authors judge Kenward-Roger "preferable" for individual effects.

Moeyaert et al. 2013 [38]. Three-level multiple-baseline synthesis with independent errors and an "external event" that shifts all participants of a study at the same occasions (constant or fading), the only time-local shock common to everyone in this group, loosely a period effect. Ignoring the event biased the effects (coverage of the trend effect as low as 0.26); modeling it restored coverage of about 0.94 to 0.97. 400 replicates per condition; MCSE nr.

Petit-Bois et al. 2016 [39]. Three-level synthesis; errors independent, AR(1) ($\rho = -0.3$, 0.3, 0.6) or MA(1), always analyzed with AR(1) and Kenward-Roger. Fixed effects nearly unbiased, coverage 0.913 to 0.981 under correct, over- and misspecified error structures; variance components badly biased. 3,000 replicates; MCSE nr.

Declercq et al. 2019 [40]. Count outcomes from a Poisson GLMM with random intercept and random treatment effect; no autocorrelation. A misspecified

normal LMM with Satterthwaite degrees of freedom held size at 0.04 to 0.05, while the correctly specified GLMM with a Wald z test was inflated (0.07 to 0.12). 2,000 replicates; MCSE given once.

Declercq et al. 2022 [41]. One-stage three-level LMM against two-stage meta-analysis of per-case effects, z and t inference; no autocorrelation. Estimates nearly identical; coverage 0.951 to 0.964 for the one-stage model with Satterthwaite degrees of freedom against 0.912 to 0.917 for two-stage; size with 5 studies 0.03 to 0.05 against 0.09 to 0.10. 1,000 replicates; MCSE nr.

Moeyaert, Yang and Xu 2022 [42] (read through a summarizing reader). Power to detect participant-level moderators of the treatment effect with Kenward-Roger degrees of freedom and no autocorrelation; about 20 participants needed for three moderators. Conceptually closest to the biomarker interaction of papers 01 and 02, not to β_D.

Chen and Pustejovsky 2024 [43]. Two-stage meta-analysis of case- level effect sizes from Poisson AR(1) data (ρ 0, 0.2, 0.4); model-based errors against robust variance estimation via `clubSandwich` with a small-sample correction and Satterthwaite degrees of freedom, the machinery of the protocol's CR2; 4,000 replicates. Robust errors protected coverage only when the case-level estimator was unbiased (quoted: robust variance estimation "cannot correct for systematic bias in effect size estimates").

A.2.8 Single-case designs: power, randomization tests and Bayesian analysis **Bouwmeester and Jongerling 2020 [44].** Multiple-baseline AB, 2 to 12 participants, 15 to 60 measurements; AR(1) ρ 0 to 0.5; **gradual onset** of the effect over 0, 1/4, 1/3 or 1/2 of the intervention phase; a one-tailed randomization test with 400 sampled permutations. 500 replicates per estimate, each scenario repeated 100 times, with the SD of power reported. Size 0.046 to 0.05 with 3 or more participants. Gradual onset cut power sharply: 0.97 with immediate onset, 0.40, 0.26 and 0.15 with onset over 1/4, 1/3 and 1/2 of the phase; $\rho = 0.5$ cut it to 0.80. The authors recommend excluding measurements taken while the effect is emerging.

Michiels et al. 2020 [45]. Randomized multiple-baseline, 3 to 6 cases; normal, uniform, AR(1) ($\rho = 0.6$) or bimodal data; a two-level LMM with random intercept and random treatment effect and Kenward-Roger F, against the same model wrapped in a randomization test with 1,000 assignments. 1,000 replicates; MCSE nr. The wrapper held the nominal level throughout; Kenward-Roger was grossly inflated for bimodal data. The text read is internally inconsistent about the model's size under the other distributions.

Levin, Ferron and Kratochwill 2012 [46] (abstract only). Randomization and permutation tests for ABAB…AB and alternating- treatment designs, systematic or randomized phase order, with or without positive autocorrelation; systematic designs kept size at or below nominal with appropriately conceived tests.

Tueller et al. 2023 [47]. AB designs with 20 to 100 participants and 30 to 100 observations; phase jump or gradual linear change; random intercept and slope, **independent** errors (quoted: "which reflects no autocorrelation among residuals"); carryover not simulated. Power only (one-sided), 1,000 replicates per condition with 95% intervals for power; size not evaluated.

Natesan Batley and Hedges 2021 [48]. Single-case AB series, phase length 5 to 15, AR(1) ρ 0 to 0.8; four Bayesian models; 100 datasets per condition. In short series, slope and AR(1) could not be separated (both credible intervals contained 0 in 66% of datasets); the authors favor simpler models.

Joo and Ferron 2019 [49] (abstract only). Non-normal multiple- baseline data; within- and between-series estimates by REML and MCMC; minimal bias, coverage "frequently acceptable, but varied".

A.2.9 The ancestor of the refined process Hendrickson et al. 2020 [4].

Designs OL, OL+BDC, CO and Hybrid over 20 weeks with 2.5 weeks of titration at each start. DGP: the three-component construction with modified Gompertz means and variances, expectancy 1 open label and 0.5 blinded, exponential decay of the last on-drug BR after discontinuation, a joint multivariate normal draw with within-factor autocorrelation, cross-factor correlations and a biomarker-BR correlation (covariance moderation), coerced to positive definite; outcome-dependent dropout. Estimand: the biomarker interaction; the paper states that its power "does not address whether the treatment is effective as compared with placebo", and **no main-effect results are reported**. Analysis: `lmer` with a random intercept and no serial-correlation or carryover term (the text also calls it an MMRM with unstructured covariance, which is inconsistent); 1,000 replicates; MCSE nr. Even a 0.1-week half-life gave "a precipitous decline in power" for the Hybrid and OL+BDC designs.

A.2.10 Screened and kept as background

- **Applications with competing analyses, no simulation.** Zucker et al. 1997 [50] (hierarchical Bayesian aggregation and shrinkage, fibromyalgia); Zucker, Ruthazer and Schmid 2010 [3] (summary and IPD meta-analysis, LMMs with four residual structures chosen by BIC, Bayesian models, and design-mimicking subsets on 58 trials; the only mention of simulation is a call for it; a precision table for patients versus period pairs). The paper 04 text that says this study "characterized the within- and between-patient variance structure" is accurate; it should not be cited as a simulation.
- **Protocols, reviews and trial reports with a single simulation.** Stunnenberg et al. 2015 [51] (a Bayesian expected-posterior- probability calculation, 1,000 iterations); Stunnenberg et al. 2018 [52] (a post hoc simulation showing power 69% and bias −0.46 on an assumed effect of

1.75 under the trial's response-based individual stopping rule); Nikles et al. 2011 [53] (one sentence on a SAS simulation for sample size under cycle-wise attrition).

- **Tutorial.** Schmid and Yang 2022 [54]: Bayesian models with trend, AR(1) terms, carryover decay regressors and missing-data imputation; no simulation of operating characteristics.
- **One simulated trajectory.** Defelippe et al. 2026 [55]: Bayesian continuous monitoring of a Poisson outcome with a placebo level shift; no replication.
- **Individualization, not the population mean.** Diaz 2021 [56] (partial empirical Bayes prediction for new patients, correlated random effects, no carryover or period effects); Cheung and Mitsumoto 2022 [58] (abstract only: analytic power for evaluating a personalized trial program).
- **Carryover as target in a parallel trial.** Sturdevant and Lumley 2021 [57]: linear post-discontinuation carryover in a prevention trial, MAR dropout by diagnosis.
- **Not read in full.** Moeyaert et al. 2013 [59] (abstract only: a three-level synthesis needing 30 or more studies and 20 or more occasions); Xie and Yu 2017 [60] (record only); Swan and Pustejovsky 2018 [61] (a gradual-effects model for reversal designs, found late; record only, high priority to read, because its build-up and dissipation of effects resembles the refined process's onset and carryover).

A.3 Comparison tables

Abbreviations: OL, open label; BDC, blinded discontinuation; CO, AB/BA crossover; MBD, multiple-baseline design; RI, random intercept; RS, random slope; CS, compound symmetry; KR, Kenward-Roger; Satt., Satterthwaite; LRT, likelihood-ratio test; MA, meta-analysis; HTE, treatment heterogeneity (τ^2); nr, not reported in what was read; "intercept only" means the source's "heterogeneity" is a random intercept, not τ^2.

Table 10: Prior studies compared with this protocol: designs and data-generating features

Study	Design	Carryover in DGP	Serial correlation	τ^2 (HTE)	Expectancy or trend
Senn [2]	aggregated, k paired cycles	assumed absent (washout)	none	yes, normal	none
Araujo [11]	aggregated, 3 cycles	none	none	yes, 0 to 1	cycle effect

Study	Design	Carryover in DGP	Serial correlation	τ^2 (HTE)	Expectancy or trend
Yang [16]	aggregated, periods with repeats	assumed absent	AR(1), CS	yes (RS)	none
Hedges [17, 18]	multi-case $(AB)^k$	none	AR(1), known	no (intercept only)	none
Shi, Ye [19]	CO vs parallel	additive shift, signed	none	via covariate interaction	period change
Blackston [12]	3 cycles vs CO vs parallel	constant added after treatment, also under null	none	no (intercept only); selection mixture	none
Carrozzo [20]	3 cycles vs CO vs parallel	10% of effect, zero under null	none	no (intercept only)	none
Chen, Chen [21]	3 to 4 cycles, 1 obs per period	0 or 20% of previous effect	CS 0 to 0.8, AR(1) 0.5, 0.8	no	none
Gärtner [22]	2 blocks x 2 periods, daily	wash-in and wash-out recursion	drift only	no	Wiener drift
Krone [23]	aggregated, random order	none	none	yes	none
Tang, Landes [24]	single person	none	AR(1) -0.33 to 0.67	n/a	rate-change slope
Wang, Schork [25]	single person, 400 obs	none (main runs)	AR(1)	n/a	none
Lin [26]	single person, 28 obs	none	AR(1) 0 to 0.7	n/a	none
Percha [27]	single person, blocks	first-order wash-in and wash-out	drift plus mean reversion	n/a	Wiener drift

Study	Design	Carryover in DGP	Serial correlation	τ^2 (HTE)	Expectancy or trend
Schork [28]	single person, gaps	none	AR(1) -0.99 to 0.99	n/a	slope (one study)
Liao [29]	single person, daily	distributed lag, 5 shapes	AR(p), ARMA	n/a	trend, weekend
Selukar [30]	single person, blocks; series	assumed absent	block intercept	no	assumed absent
Jiang [31]	single person, cycles	assumed absent (washout)	none	three-point mixture	none
Senarathne [32]	aggregated, 3 cycles	none	none	yes	none
Shrestha [33]	individual and aggregated	nr (abstract)	nr	nr	nr
Giai [34]	series, 38 patients	nr (abstract)	nr	nr	nr
Cruz [35]	CO, repeated measures	mean-zero functions	none	no	within-period trend
Ferron [36, 37]	MBD, 4 to 8 cases	none	AR(1) 0 to 0.4	yes, 0.1 to 0.5	none
Moeyaert [38]	MBD, 3 levels	none	none	yes	external event
Petit-Bois [39]	MBD, 3 levels	none	AR(1), MA(1)	yes	slope change
Declercq [40]	MBD, counts	none	none	yes	none
Declercq [41]	AB, 3 levels	none	none	yes	linear, quadratic
Moeyaert [42]	MBD	none	none	yes, moderated	none
Chen, Pustejovsky [43]	AB series, counts	none	Poisson AR(1)	study and case level	none
Bouwmeester [44]	MBD, 2 to 12	none; gradual onset	AR(1) 0 to 0.5	no	none

Study	Design	Carryover in DGP	Serial correlation	τ^2 (HTE)	Expectancy or trend
Michiels [45]	randomized MBD	none	AR(1) 0.6 (one level)	per-occasion noise	none
Levin [46]	ABAB, alternating	nr (abstract)	0 or positive	nr	nr
Tueller [47]	AB, 20 to 100	none	none	yes (RS)	gradual slope
Natesan Batley [48]	single-case AB	none	AR(1) 0 to 0.8	n/a	slopes
Joo, Ferron [49]	MBD, non-normal	nr (abstract)	nr	nr	nr
Hendrickson [4]	OL, OL+BDC, CO, Hybrid	exponential decay of BR	within-factor autocorrelation	via BR variance	expectancy 1 / 0.5; trend
This protocol	Hybrid, OL+BDC, CO, RCT (multi-crossover optional)	anchored exponential, $t_{1/2}$ 0 to 0.2 wk; behavioral 1 wk; Weibull	separable AR(1), $\rho = 0.7$ per week	yes, τ 0 to 5.49, heterogeneous nulls	η_t 1 / 0.5; Gompertz TV

Table 11: Prior studies compared with this protocol: analyses, inference and performance measures

Study	Analyses compared	Inference, df	Measures	MCSE	Reps per cell	Comparator
Senn [2]	summary-measures t; fixed-effects ANOVA; shrinkage	$N-1$ vs $N(k-1)$	analytic power, SE	n/a	n/a	none

Study	Analyses compared	Inference, df	Measures	MCSE	Reps per cell	Comparator
Araujo [11]	paired t, mixed models, summary measures, MA, BLUP	df decompositions	empirical vs model variance	no	1,000	none
Yang [16]	LMM with RI/RS	GLS, known variances, z	analytic power	n/a	n/a	none
Hedges [17, 18]	between-case standardized contrast	Box df, known ρ	power ([17]); bias ([18])	no	8,000 ([18])	none
Shi, Ye [19]	crossover and parallel z, covariate-adjusted	asymptotic	size, power	no	2,000 to 10,000	parallel
Blackston [12]	`lmer` RI, LRT; OLS	asymptotic LRT	size, power	no	5,000	parallel, CO
Carrozzo [20]	t tests; two-stage MA	z interval	size, power	no	5,000	parallel, CO
Chen, Chen [21]	paired t; 2 mixed models; DerSimonian-Laird	nr	size, power, bias, MSE	no	3,000	single person

Study	Analyses compared	Inference, df	Measures	MCSE	Reps per cell	Comparator
Gärtner [22]	OLS, decay-regressor OLS, Bayesian networks, G-estimation	empirical SD	bias, SE	no	100 subsamples	none
Krone [23]	two-stage, LMM, Bayesian LMM	none	RMSE of individual effects	no	1,000	none
Tang, Landes [24]	serial vs usual t	effective df	size, power, interval width	target	10,000	none
Wang, Schork [25]	GLS AR(1) LRT	asymptotic	power, size	no	1,000	none
Lin [26]	GLS AR(1) Wald vs bootstrap	normal ref.; bootstrap	size, power	tolerance band	10,000	none
Percha [27]	OLS with blocks; median rule	OLS	power, estimate	no	50	none
Schork [28]	OLS, GLS, HAC, SPRT	t, HAC	false positives, power	partial	100 to 10,000	none
Liao [29]	Bayesian lag models	posterior	bias, RMSE	no	100	none

Study	Analyses compared	Inference, df	Measures	MCSE	Reps per cell	Comparator
Selukar [30]	LMM with boundaries; combination	normal vs t boundaries	size, power, bias, MSE	no	10,000	fixed design
Jiang [31]	group-sequential mean difference	known variance	decision rates, sample size	no	1,000	fixed design
Senarathne [32]	adaptive vs fixed allocation, Bayesian GLMM	posterior	precision, correct choice	no	20	fixed design
Shrestha [33]	Bayesian hierarchical bandit	posterior	nr (abstract)	nr	nr	nr
Giai [34]	generalized pairwise comparisons	nr	size, power (abstract)	nr	nr	none
Cruz [35]	GEE with and without carryover splines	sandwich, no correction	bias, coverage	no	500	none
Ferron [36]	LMM, independent vs AR(1)	containment, residual, between-within, Satt., KR	t; coverage, width; τ^2 bias	single value	5,000	none

Study	Analyses compared	Inference, df	Measures	MCSE	Reps per cell	Comparator
Ferron [37]	per-person OLS vs empirical Bayes	containment, Satt., KR	individual bias, MSE, coverage	single value	1,000	none
Moeyaert [38]	two-stage three-level, with and without event terms	Satt.	bias, coverage	no	400	none
Petit-Bois [39]	three-level LMM with AR(1)	KR	bias, coverage	no	3,000	none
Declercq [40]	LMM vs Poisson GLMM	Satt. vs Wald z	size, power, bias	single value	2,000	none
Declercq [41]	one-stage vs two-stage	Satt. t; z	size, coverage, bias	no	1,000	none
Moeyaert [42]	HLM with moderators	KR	power	no	1,000	none
Chen, Pustejovsky [43]	multilevel MA, robust variance	CR2-type, Satt.	bias, RMSE, coverage	no	4,000	none
Bouwmeester [44]	randomization test	permutations	size, power	SD of power	500 x 100	none
Michiels [45]	LMM KR vs randomization test wrapper	KR; permutation	size, power	no	1,000	none

Study	Analyses compared	Inference, df	Measures	MCSE	Reps per cell	Comparator
Levin [46]	randomization and permutation tests	permutations	size, power (abstract)	nr	nr	none
Tueller [47]	LMM RI+RS, piecewise	nr, one-sided	power only	95% intervals	1,000	none
Natesan Batley [48]	four Bayesian models	posterior	RMSE, bias, coverage	no	100	none
Joo, Ferron [49]	REML vs MCMC, two estimators	nr	bias, coverage (abstract)	nr	nr	none
Hendrickson [4]	`lmer` RI (interaction only)	nr	power, bias	no	1,000	among four designs
This protocol	19 analyses (Section 5.2) incl. RS, summary measures, oracles	CR2 Satt.; KR; containment; exact t; HC2	size, size-adjusted power, bias, empirical SE, κ, coverage, τ^2_d, BLUP RMSE	every measure	1,000 to 2,000 null; 500 to 1,000 alt.	parallel RCT (ANCOVA, MMRM)

A.4 Evaluation

What the literature does well (inspected unless marked).

- **The variance algebra is settled.** Senn [2] and Araujo et al. [11] give the variance of the mean effect with τ^2, separate the strict null (no effect in anyone) from the population null (mean zero), and show that an analysis that ignores τ^2 is valid for the first and anticonservative for the

second. This is the single most useful result for the protocol, because its heterogeneous nulls (block 2) are exactly the population null.

- **Small-sample inference in LMMs with AR(1) errors has been studied carefully,** but in multiple-baseline designs. Ferron et al. [36] compared five degrees-of-freedom methods with AR(1) modeled or not; Satterthwaite and Kenward-Roger gave coverage near 0.95 only when AR(1) was modeled, and containment was conservative. Declercq et al. [41] and Selukar et al. [30] show that the choice between normal and t references changes size materially at small N or few blocks.
- **Randomization tests** give exact size under the design's own randomization and are robust to non-normal data (Michiels et al. [45], Bouwmeester and Jongerling [44], Levin et al. [46]).
- **Onset and carryover have mechanistic models** outside the aggregated-trial literature: first-order wash-in and wash-out (Percha et al. [27], adopted by Gärtner et al. [22]), distributed lags with non-monotone shapes (Liao et al. [29]), gradual onset with a quantified power cost (Bouwmeester and Jongerling [44]).
- **Individual effects and their intervals** are evaluated (Ferron et al. [37], Krone et al. [23], Senn [2]); Bayesian hierarchical and adaptive designs exist ([32], [33], [50]); count and ordinal outcomes are covered in the single-case literature ([40], [43]).

Where a DGP or analysis would mislead in a Hybrid, titrated-drug setting (argued, from the features in Table "Prior studies compared with this protocol: designs and data-generating features").

- **Constant, immediate effect.** Every aggregated-trial simulation found uses a constant effect from the first on-drug observation. Under re-titration a one-week exposure reaches 3.1% of the ceiling (Section 3.3), so a constant-effect DGP overstates the information in short re-exposures, and its power comparisons with a parallel trial favor the N-of-1 design more than a titrated drug would. Bouwmeester and Jongerling's onset result [44] (power from 0.97 to 0.40 when the effect builds over a quarter of the phase) shows the size of the effect in a related design.
- **Carryover that is constant, mean-zero or zero under the null.** Blackston et al. [12] add carryover even when there is no effect, so their size inflation (up to 27%) measures a carryover with no drug behind it; Carrozzo et al. [20] make carryover proportional to the effect, so it cannot affect size; Cruz et al. [35] center it to mean zero, so it cannot cause bias. In the refined process carryover is the decayed drug response, zero under the homogeneous null and nonzero for individuals under the heterogeneous null (derived from Section 3.2). None of these three choices reproduces that.
- **Carryover excluded with oracle knowledge, or ignored.** Chen and Chen [21] adjust with the true carryover indicators; the v1 paper 04 driver

drops the rows the DGP flags (Section 1, problem 4). Both remove the problem the design must solve.

- **No expectancy.** No study other than Hendrickson et al. [4] has an expectancy component that differs between open-label and blinded phases; only Defelippe et al. [55] has a placebo level shift. The drug-expectancy confounding of the Hybrid and OL+BDC designs (Glossary) is therefore untested in the main-effect literature.
- **No treatment heterogeneity, under a name that suggests it.** Blackston et al. [12], Carrozzo et al. [20] and Hedges et al. [17, 18] vary an intercept variance that cancels in within-person contrasts and call it heterogeneity or write it τ^2. Their comparisons with parallel trials therefore show the gain from removing between-person baseline variance, not the cost of τ^2, which in Senn's variance formula [2] becomes the dominant term as k grows (derived).
- **Comparisons that are not information-matched.** Blackston et al. [12] and Carrozzo et al. [20] compare 6 observations per person with 2 (crossover) and 1 (parallel). The power advantage they report combines the design's within-person contrast with three to six times as many measurements.
- **Inference.** Asymptotic likelihood-ratio or Wald tests with normal references (Blackston et al. [12], Wang and Schork [25], Lin et al. [26], Carrozzo et al. [20]), unstated degrees of freedom (Chen and Chen [21], Tueller et al. [47]), and sandwich errors without small-sample correction at 2 to 10 clusters (Cruz et al. [35]). Lin et al. [26] show Wald size up to 0.113 at 28 observations; the protocol's CR2 with Satterthwaite degrees of freedom is the kind of correction this literature lacks.
- **MCSE.** Of 38 studies, 8 report a Monte Carlo error in some form ([24], [26], [28], [36], [37], [40], [44], [47]), most as a single value or a band for one measure; none reports it for every measure. Several have 20 to 100 replicates per cell ([22], [27], [29], [32], [48]), too few to resolve size differences of 0.01 to 0.02.

What is new in this protocol relative to the literature found (argued; "new" means not found in the 38 studies, not a claim about all published work).

- A main-effect simulation of the Hybrid and OL+BDC designs. The only study of these designs, Hendrickson et al. [4], targets the interaction and reports no main-effect result.
- An expectancy component that changes between open-label and blinded phases, and an analysis that models it (F1), so that drug-expectancy confounding appears as bias rather than being built into the target.
- Gompertz onset with restart at re-exposure, combined with anchored exponential carryover and a design-specific estimand (β_D^{exp}, Section 4) whose value is computed per cell.
- Heterogeneous nulls with τ set in absolute units, and a heterogeneity that follows the drug-response profile (zero before exposure, decaying after it).

- An RCT comparator generated from the same component model with a baseline measured with error, and an all-visits MMRM comparator (AM).
- Carryover handled without oracle information (XS, D_{bc} at a misspecified half-life) and the oracle kept only as a labeled bridge.
- Closed-form probability limits and sandwich variances before simulation, with the simulation validating them (Section 5.1).
- MCSE targets and MCSEs for every measure, size-adjusted power and a working decision rule with an estimation screen.

What the literature handles better, or the protocol lacks (argued).

- **Randomization tests.** The protocol has none, although its paths are randomized; a permutation test is the one test whose size does not depend on the covariance model (Michiels et al. [45]).
- **Bayesian hierarchical analysis** of τ^2 and individual effects ([50], [32], [33], [54]) is absent; with 35 to 150 participants and a weakly identified slope variance, a Bayesian model with a proper prior on τ may estimate τ^2 better than REML at the boundary. Ferron et al. [36] and Petit-Bois et al. [39] show how poorly REML estimates treatment-effect variances with few cases.
- **Individual-effect inference.** The protocol reports BLUP RMSE and correlation (Section 6.1) but not interval coverage for individual effects, and has no per-participant OLS comparator; Ferron et al. [37] show the degrees-of-freedom choice matters more for these intervals than for the mean.
- **Design adaptivity and sequential monitoring** ([30], [31], [32], [33]) are out of scope here; the bias from response-based individual stopping in a real aggregated trial (Stunnenberg et al. [52]) is a reason to keep stopping rules out of the rebuilt study's DGP unless they are studied deliberately.
- **Count, ordinal and bounded outcomes** ([40], [43], [22], [27]): the protocol's outcome is a normal total score with no range check (`docs/51`, Feature 17).
- **Non-monotone or slow-absorption carryover** (Liao et al. [29]) and **sign-dependent carryover** (Shi and Ye [19]: positive carryover attenuates and is conservative, negative carryover inflates size). The refined process has only positive, monotone carryover; a rebound after discontinuation is not represented.
- **Common time shocks** (Moeyaert et al. [38]) and **non-representative samples** (Blackston et al. [12], selection mixture): neither is in the protocol; the first is partly covered by the visit-factor analyses, the second is outside the study's aims.

Corrections to the current paper 04's literature summary (inspected against the sources; for Section 9's rewrite):

- Chen and Chen [21] has two authors; "Chen, Chen and Xia" is wrong (the bib key problem of Section 9 is separate).
- Blackston et al. [12] and Carrozzo et al. [20] vary intercept variance, not treatment heterogeneity; "patient heterogeneity" in paper 04 should say so.
- The washout finding attributed to Wang and Schork [25] is not in the text read.
- Selukar et al. [30]: nominal size is reached with t-adjusted boundaries, not "with more blocks or alpha-spending corrections".
- Tang and Landes [24]: the size values quoted in paper 04 come from figures not read here and must be checked before reuse.

A.5 Implications for the protocol

Each item is a proposal for the user's decision. None is applied to Sections 1 to 7; all are listed in Section 8 as items 12 to 20.

- **L1. Add a randomization test** (Section 8, item 12). Re-draw the path labels under the protocol's allocation rule (uniform, at least 2 per path) and recompute the statistic. For PR, a closed-form linear statistic, 999 permutations per trial cost little. For F1, which needs a refit per permutation, run 199 permutations in the cells of blocks 1 and 2 only. Reason: exact size under the strict null ([45], [44]); a reference against which CR2 and KR size can be read. Caveat (derived): a permutation test of path labels tests the strict null of no effect in anyone, not the population null of the heterogeneous cells, so its size there is itself a result.
- **L2. Add a Bayesian random-slope analysis, or record its omission** (item 13). Normal priors on fixed effects, a half-normal or half-t prior on τ, evaluated by the same frequentist measures in blocks 1 to 3 at $N = 35$ and 70. Cost: MCMC per trial is likely ten or more times an `nlme` fit (assumed); a pilot would decide.
- **L3. Add an onset-exclusion analysis XT** (item 14): F1 with the first on-drug visit after each start or restart dropped by schedule, the onset counterpart of XS. Reason: Bouwmeester and Jongerling [44] recommend excluding observations while the effect is emerging, and under re-titration the first on-drug visits carry little signal.
- **L4. Add a literature bridge cell** (item 15): constant effect, no PB or TV, $t_{1/2} = 0$, AR(1) only, Hybrid and CO. In it PW and RS can be checked against the closed forms of Senn [2], Araujo et al. [11] and Yang et al. [16], and the protocol's code against published results. Reason: none of the published DGPs can be reproduced inside the refined process otherwise.
- **L5. Cross the heterogeneous null with carryover** (item 16): block 2's null cells at $t_{1/2} = 1$ week. Reason (derived): only there does carryover

exist under a null of the refined process, so only there can the Blackston-type size inflation [12] arise in a form consistent with pharmacology.

- **L6. Add a two-point heterogeneity cell** (item 17): v_i taking two values with the variance of the $\tau = 5.49$ cell, as a responder and non-responder population (cf. the mixtures of [12] and [31]). It extends item 9.
- **L7. Designate the RCT comparator for design comparisons** (item 18): AM, which uses every visit, as the information-matched comparator, and AN as the conventional one, so that the design comparison does not repeat the unmatched comparisons of [12] and [20].
- **L8. Add measures** (item 19): interval width for β_D ([36]); coverage of individual-effect intervals with Satterthwaite and KR degrees of freedom and a per-participant OLS comparator ([37]).
- **L9. Add a bounded-outcome sensitivity cell** (item 20): Y rounded and truncated to the CAPS range in the base alternative cell, with the fraction of out-of-range scores reported (`docs/51`, Feature 17; [40], [43]).

Proposals considered and not made: sequential monitoring and adaptive allocation ([30] to [33]), because they change the design under study; non-monotone carryover ([29]) and negative (rebound) carryover ([19]), because the refined process has no mechanism for them and adding one is a DGP decision beyond this review; common time shocks ([38]), because the visit-factor analyses already absorb any shock common to all paths at a visit.